

Advanced Cognitive Neuroscience

Week 39: Basic physiology and Evoked responses

The course plan

Week 37:

Monday: Workshop paradigm: Measuring visual subjective experience
MR Recordings (Lau)

Week 38:

Monday: What is it all about? (Lau)
Workshop 1: MR-recordings (Laura & Lau)

Week 39:

Monday: Basic physiology and Evoked responses (Lau)
Wednesday: Preprocessing and evoked responses (Laura)

Week 40:

Monday: Forward modelling and inverse modelling (Lau)
Wednesday: Source reconstruction (Laura)

Week 41:

Monday: Multivariate statistics (Laura) (Lecture)
Wednesday: Multivariate statistics (Laura) (Coding)

Deadline for feedback: Video Explainer

Week 43:

Monday: Inverse modelling: beamforming (Lau) (Lecture)
Wednesday: Inverse modelling: beamforming (Laura) (Coding)

Week 44:

Monday From single subject analysis to group analysis (Laura)
Wednesday: Work on your reports (Laura)

Week 45:

Monday: What about that other cortex? - the cerebellar one (Lau)
Wednesday: Work on your reports, continued (Laura)

Week 46:

Monday: Optically pumped magnetoencephalography (Laura)
Wednesday: Respiration (Laura)

Deadline for feedback: Lab report

Week 47:

No teaching: prepare presentations!

Week 48:

Monday: Oral presentations (part 1) (Lau)
Wednesday: Oral presentations (part 2) (Lau)

Week 49:

Monday: Oral presentations (part 3) (Lau)
Wednesday: Oral presentations (part 4) (Lau)

Week 50:

Wednesday: What was it all about (Lau)

Follow-up, Tesla

TABLE 1.1 Approximate Sizes of Different Magnetic Fields of the Environment and the Body (in units of femtotesla or 10^{-15} tesla = 10^{-15} T)

Magnetic resonance imaging	3,000,000,000,000,000 (= 3 T)
Steady magnetic field of the earth	50,000,000,000
Magnetocardiogram	100,000
Brain's alpha rhythm	1,000
Brain's evoked responses	100
Sensitivity of a magnetometer	3
Noise within a magnetically shielded room	1

[https://en.wikipedia.org/wiki/Tesla_\(unit\)](https://en.wikipedia.org/wiki/Tesla_(unit))

Follow-up, Tesla

Definition [\[edit \]](#)

A particle, carrying a charge of one [coulomb](#) (C), and moving perpendicularly through a magnetic field of one tesla, at a speed of one metre per second (m/s), experiences a force with magnitude one [newton](#) (N), according to the [Lorentz force law](#). That is,

$$T = \frac{N \cdot s}{C \cdot m}.$$

Expressed in [SI base units](#), 1 tesla is:

$$T = \frac{kg}{A \cdot s^2},$$

where A is [ampere](#), kg is [kilogram](#), and s is [second](#).^[3]

[https://en.wikipedia.org/wiki/Tesla_\(unit\)](https://en.wikipedia.org/wiki/Tesla_(unit))

Video explainer

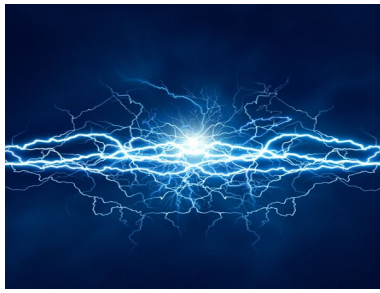
- Concepts to explain
 - post-synaptic potentials
 - current dipole and equivalent current density,
 - open field vs closed field
 - radial and tangential sources
 - volume conduction
 - evoked responses
- Deadline Thursday (23.59) in two weeks (week 41) (for feedback)

Video explainer

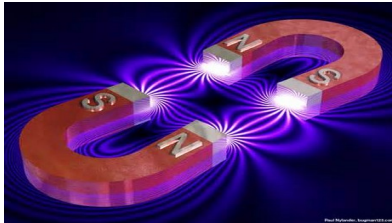
Just you talking, explaining the relevant concepts. You can gesture, but no props and no screen recordings (only I will see it)

Learning goals

- Learning
 - where the electrophysiological signals mainly originate from
 - where and how are the electrophysiological signals measured
 - how the signals measurable from radial and tangential dipoles differ



Electro·encephalo·graphy (EEG)
Magneto·encephalo·graphy (MEG)



When to use MEEG: When to use fMRI:

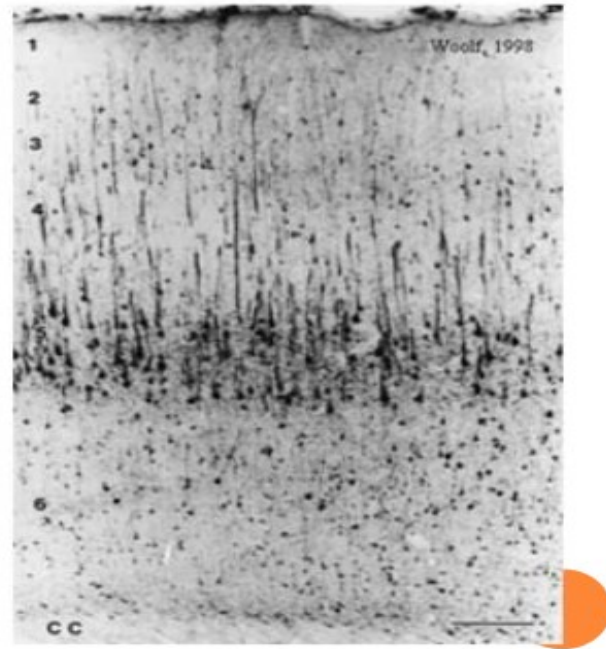
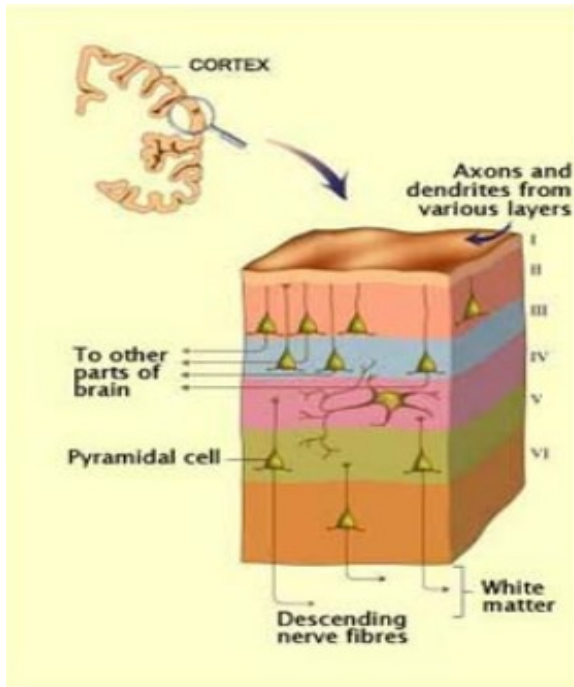
Interested in temporal aspects (< 1 ms)

Interested in *both* temporal and spatial aspects (< 1 cm)

Interested mainly in spatial aspects (< 1 mm)

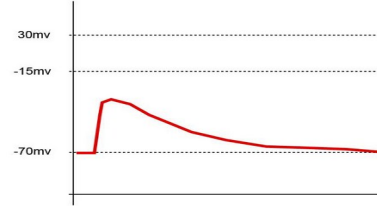
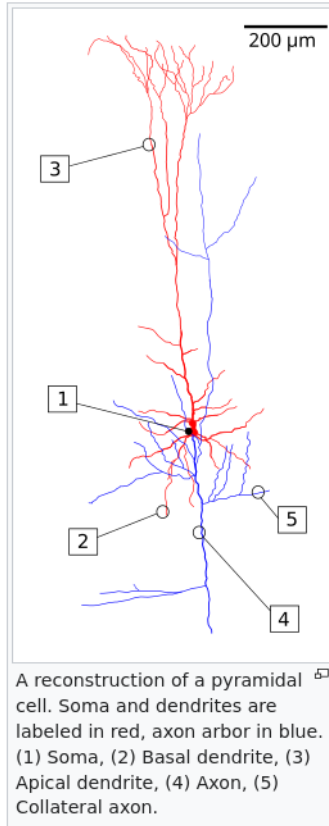
What generates the magnetic field and the electric potential?

(Primarily) pyramidal cells in the neocortex



What generates the magnetic field?

(Primarily) the excitatory post-synaptic potentials in apical dendrites in pyramidal cells in the neocortex



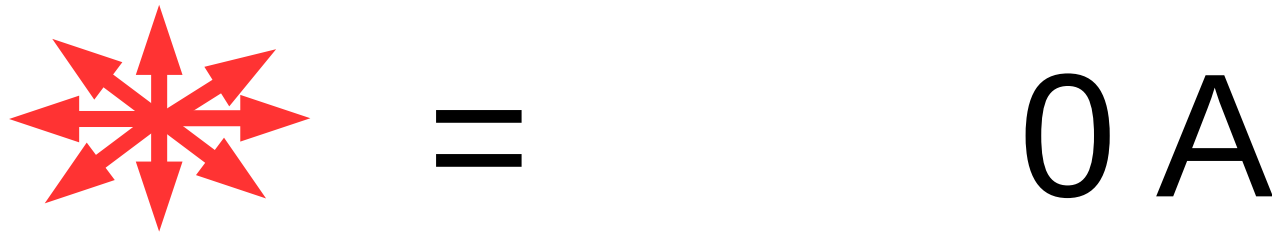
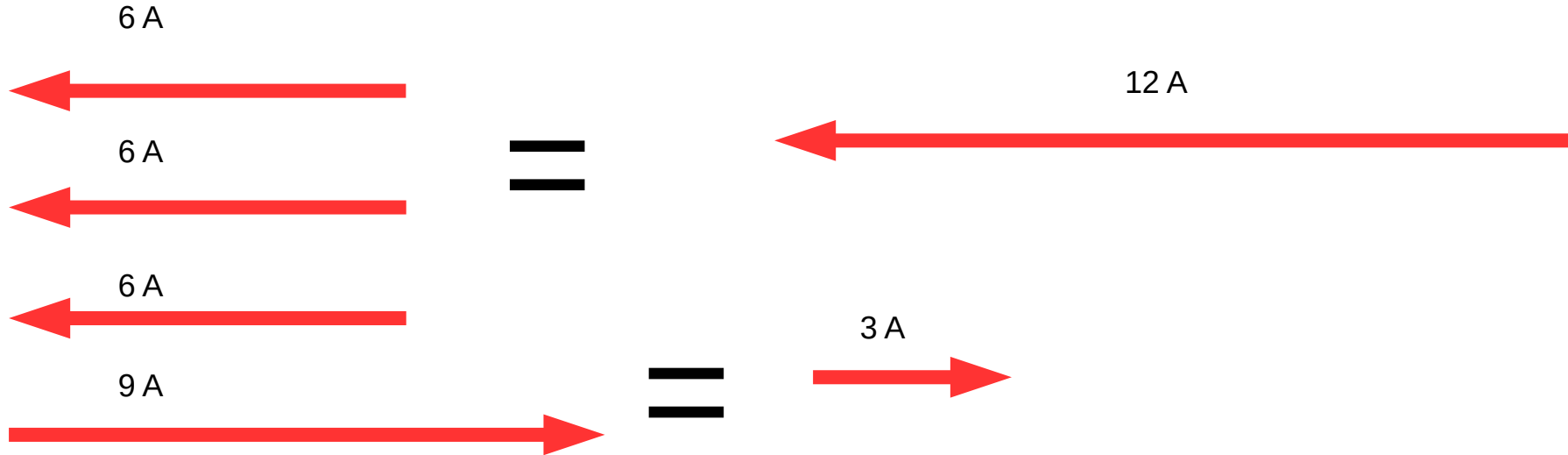
Excitatory Post-Synaptic Potential

Action potentials are too brief to sum temporally, and the axons do not sum spatially

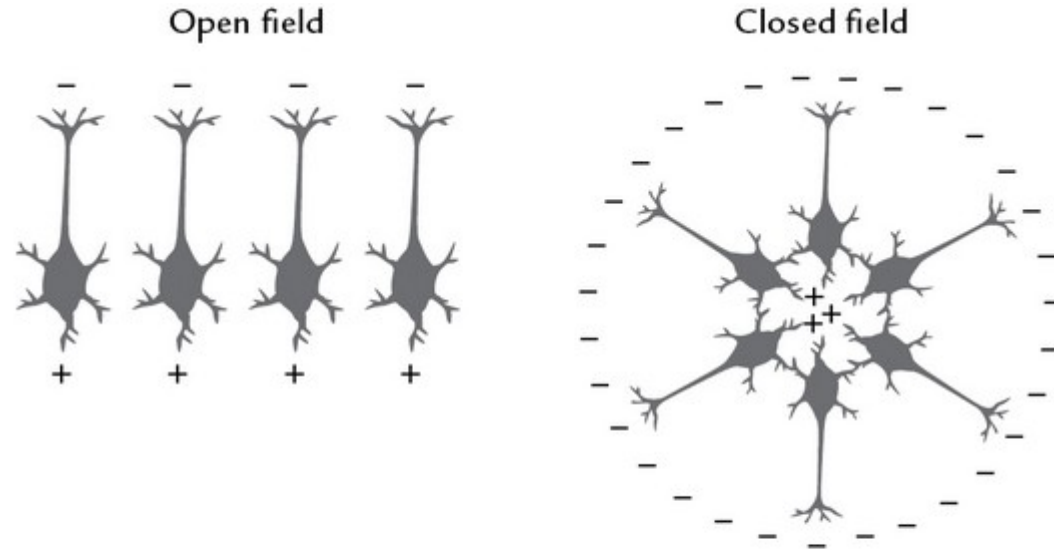
Single neurons can be seen with intracranial recordings, but generate signals many orders too weak to be seen at the scalp where we measure

Can we see the SPATIAL and TEMPORAL summations if there are “many” neurons firing at the same time?

SPATIAL Summation of currents



Open and closed fields

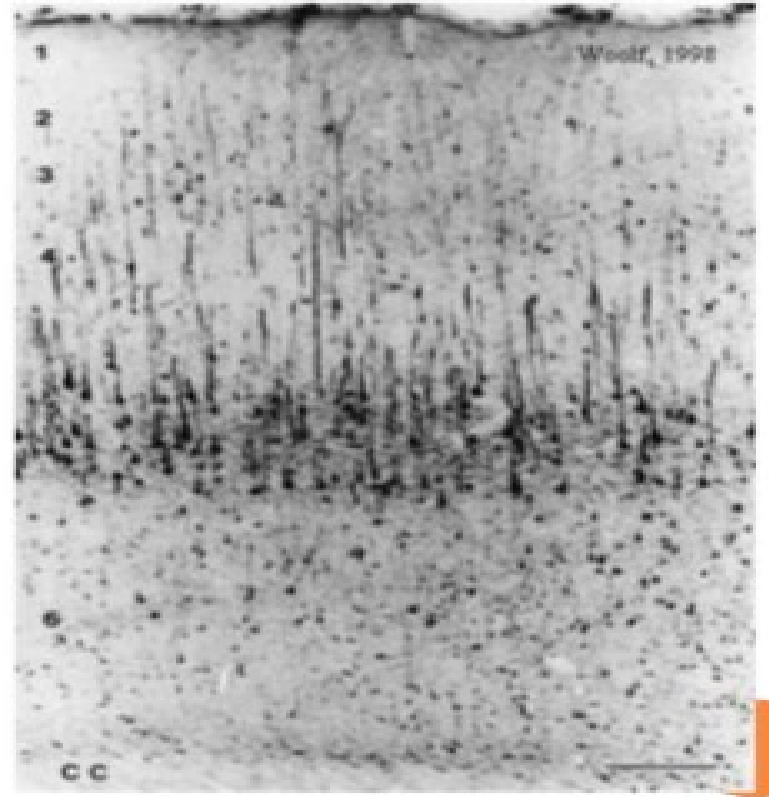


Hari and Puce, 2023

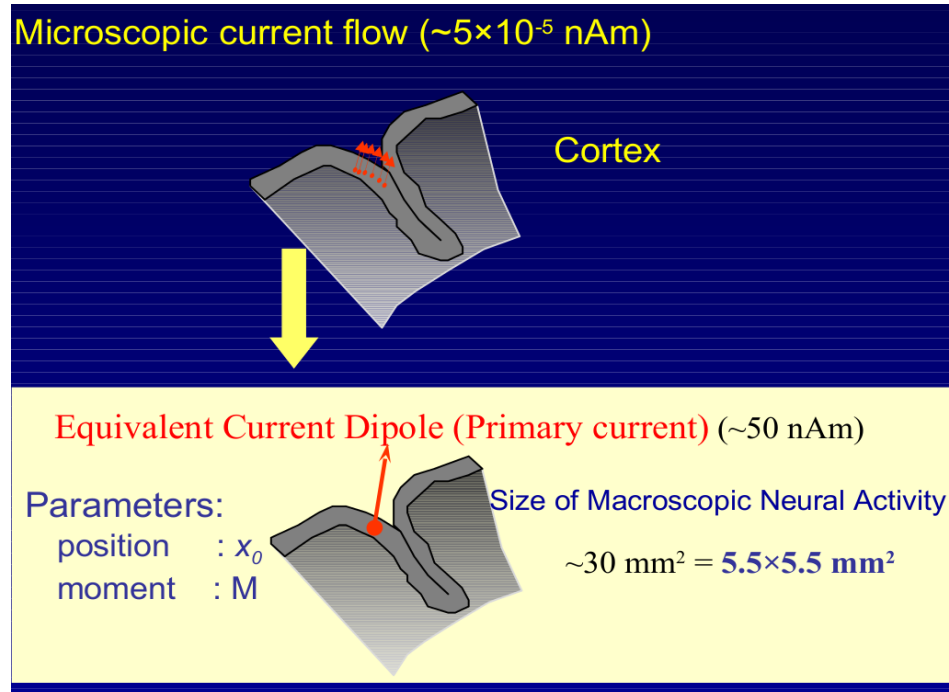
FIGURE 3.5. Open and closed fields. Polarized (activated) pyramidal cells aligned side-by-side will produce an open field whose potential and magnetic field can be seen at a distance (left panel). If the same elements are in a random order, or are radially symmetric (right panel), they will form a closed field that cannot be seen with MEG nor EEG; instead, a locally sited electrode must be inserted very close to the cluster to be able to record the potential.

SPATIAL Summation of currents

The ordering of the dendrites of pyramidal neurons creates good conditions for seeing the SPATIAL summation of post-synaptic potentials, but not the action potentials along the axons

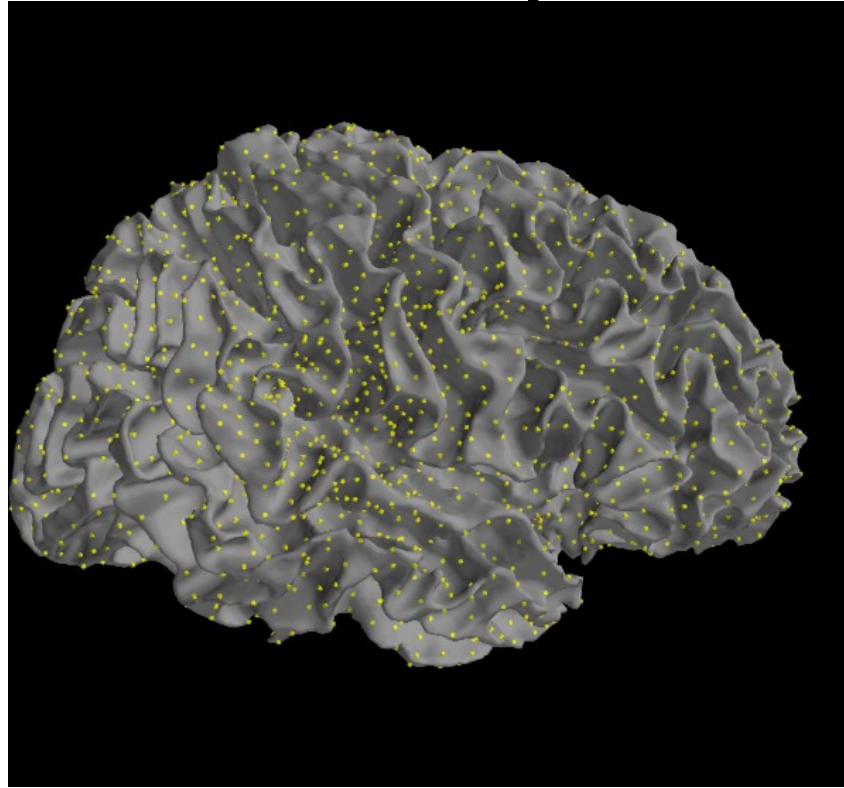


Equivalent Current Dipole



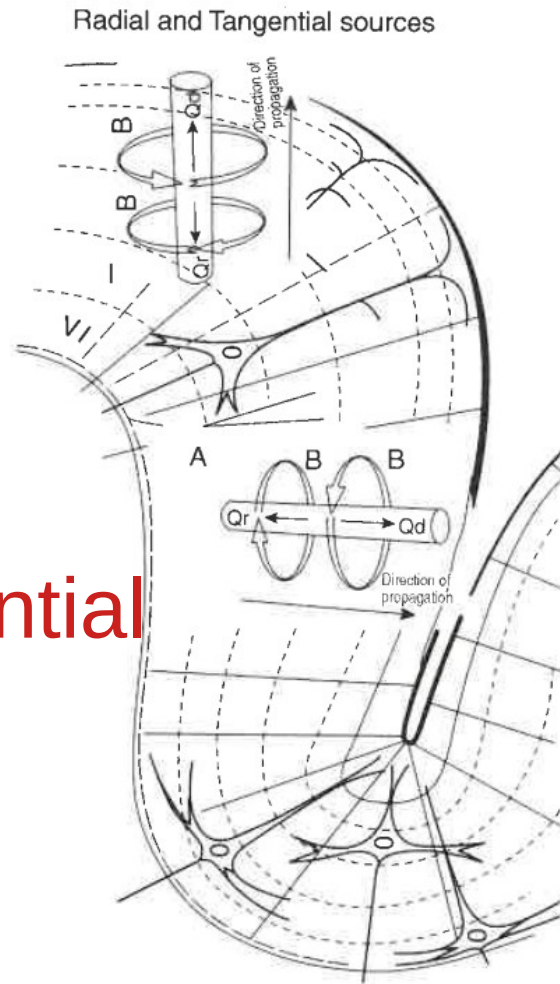
From Stephanie Sillekens

Source space with equivalent current dipoles



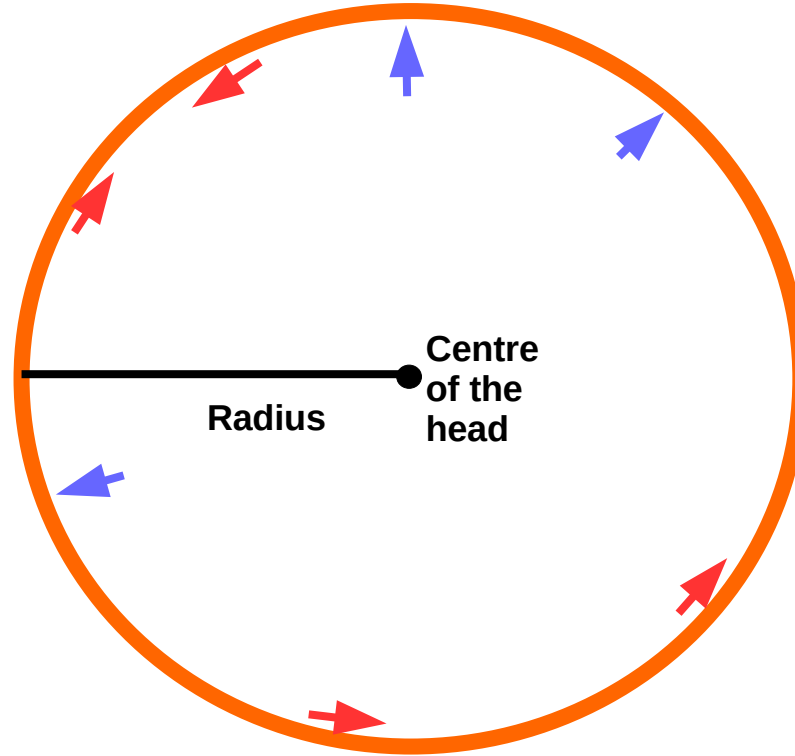
Radial

Tangential

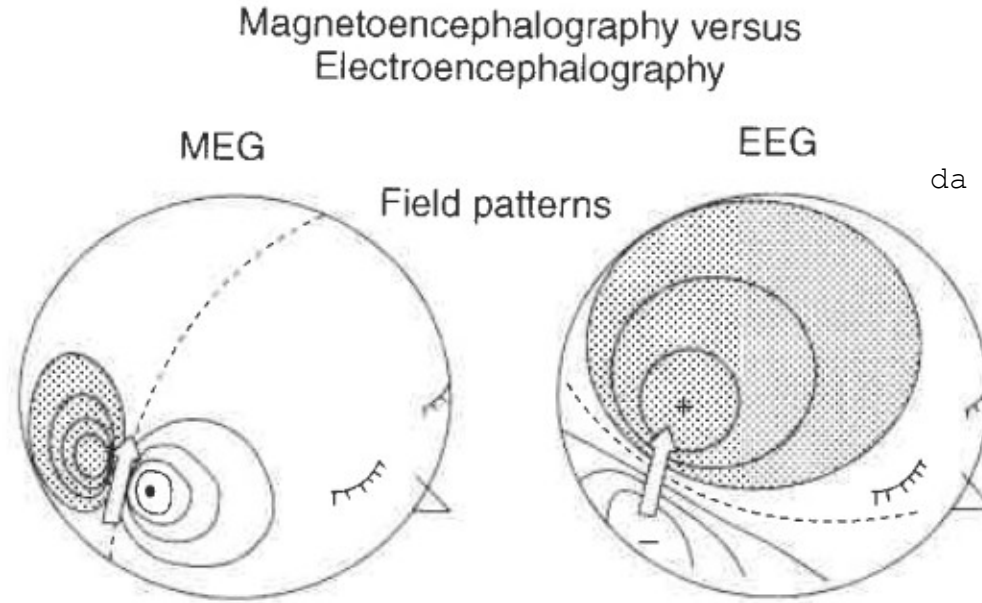


da Silva, 2010

Radial and tangential dipoles (primary current)

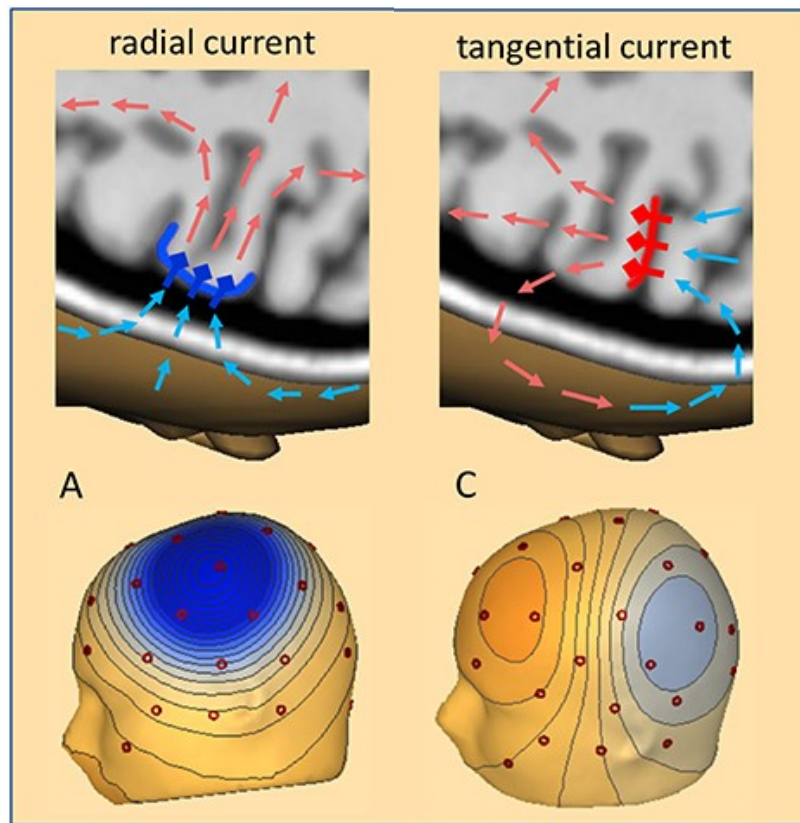


Tangential dipoles



da Silva, 2010

EEG – radial vs. tangential

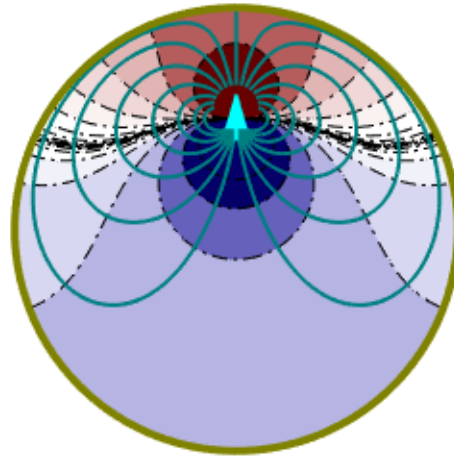


Scherg et al., 2019

MEG – what about radial sources?

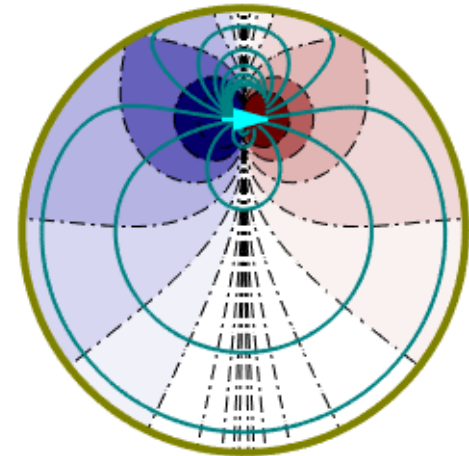
The magnetic field (B) is proportional to the vector sum of all electric current (Q_T). For a radial dipole, the primary current (Q_p) and the volume currents (Q_v) *cancel* one another ($Q_p + Q_v = Q_T = 0 \Rightarrow B = 0$) due to the perfect *symmetry* of the volume current in a sphere-shaped volume conductor

Radial



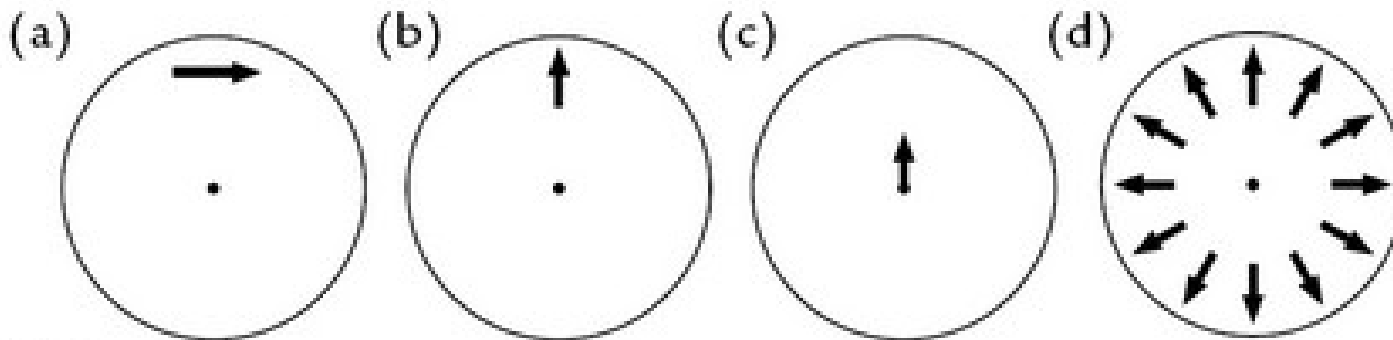
Courtesy of Christopher Bailey
https://github.com/cjayb/meeg_training

Tangential



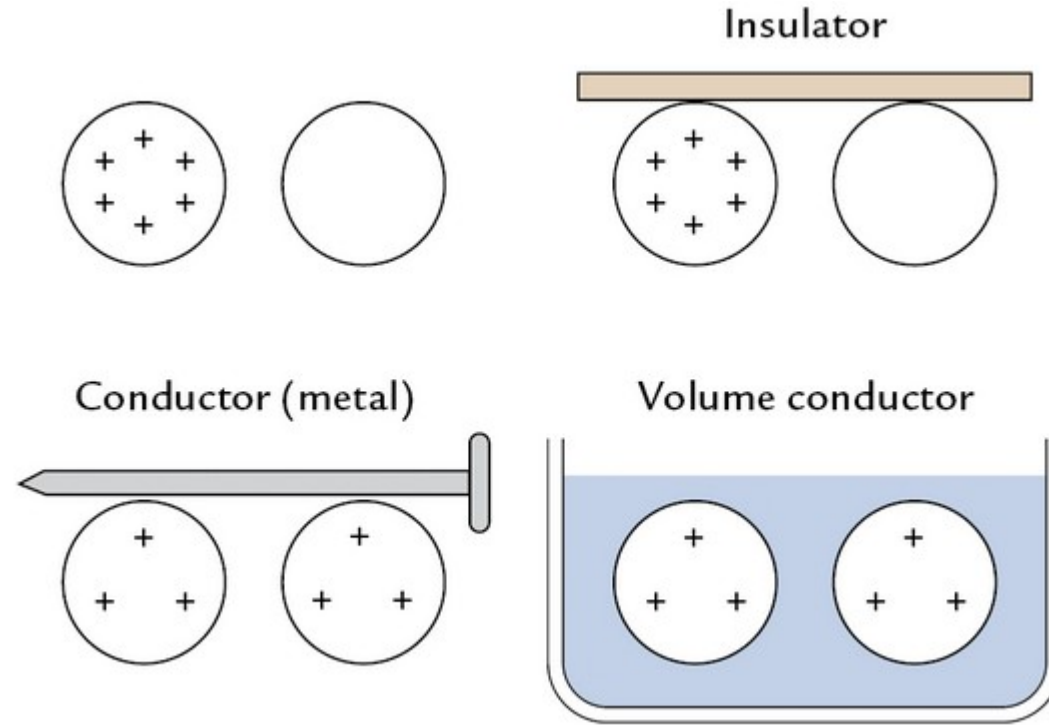
For a tangential dipole, the volume currents are *asymmetric*, thus $Q_T \neq 0 \Rightarrow B \neq 0$

Quiz



Hari and Puce, 2023

Volume conductor



Volume conductor (head model)

```
## BOUNDARY ELEMENT METHOD

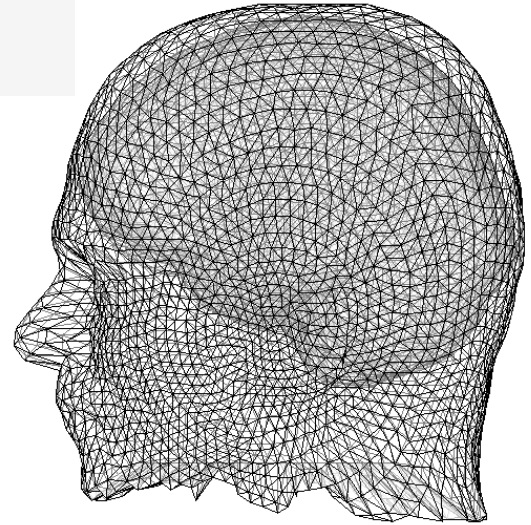
## describe the surfaces and their conductivities
#bem_model = mne.bem.make_bem_model(subject=subject, subjects_dir=subjects_dir,
#                                   conductivity=[0.3, 0.006, 0.3]) ## three layer model

## model how electrical potentials spread to the electrodes and how the
## currents of the brain are related to the magnetic field measured at the
## sensors
#bem_solution = mne.bem.make_bem_solution(bem_model)

bem_solution = mne.bem.read_bem_solution('../subjects/sample/bem/sample-5120-5120-5120-bem-sol.fif')

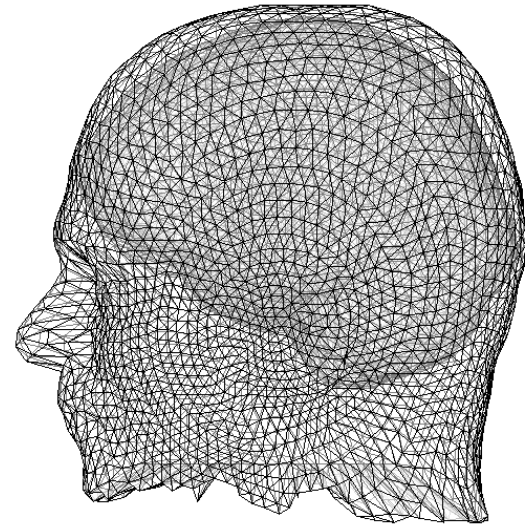
print(bem_solution['solution'])
```

An anatomical model that models the conductivities of different tissues *bem*



Volume conductor (head model)

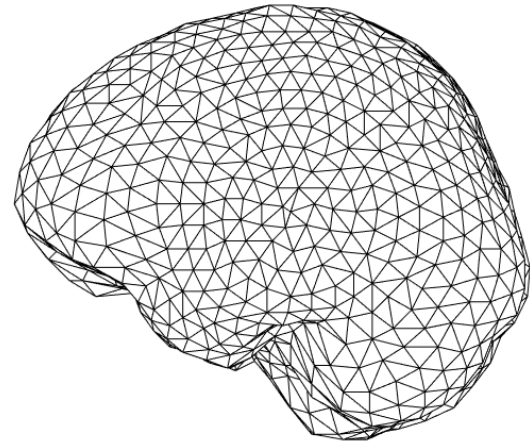
For *EEG*, we need to model the brain, skull and scalp with different conductivities (we are measuring volume currents)



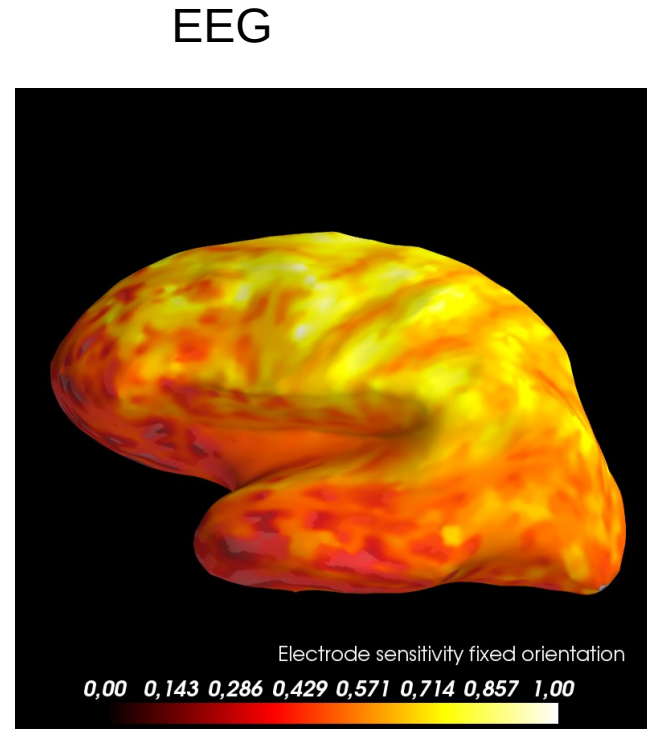
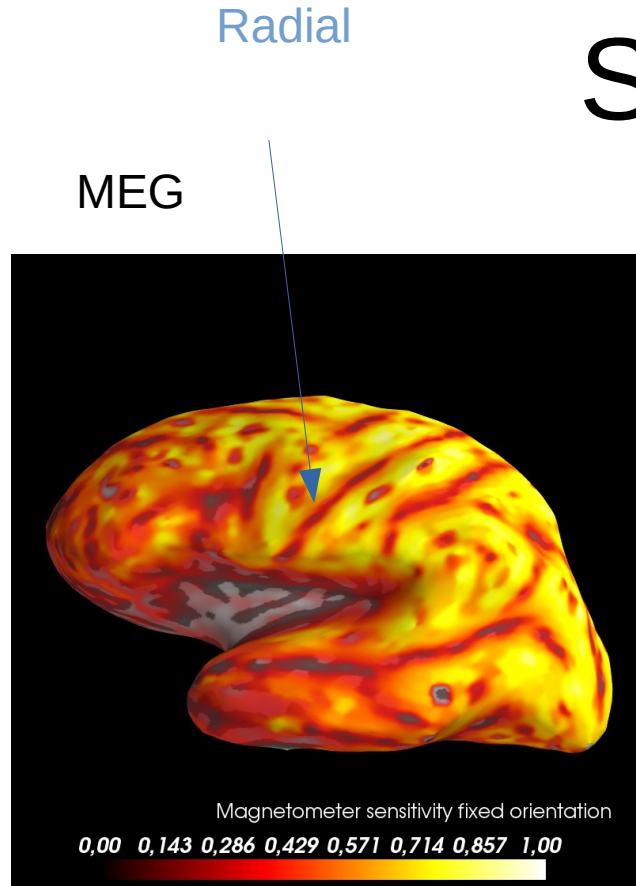
Volume conductor (head model)

For MEG, the model can be much simpler since the magnetic field is roughly proportional to the magnitude of the primary current

(This is why MEG has better spatial resolution than EEG)

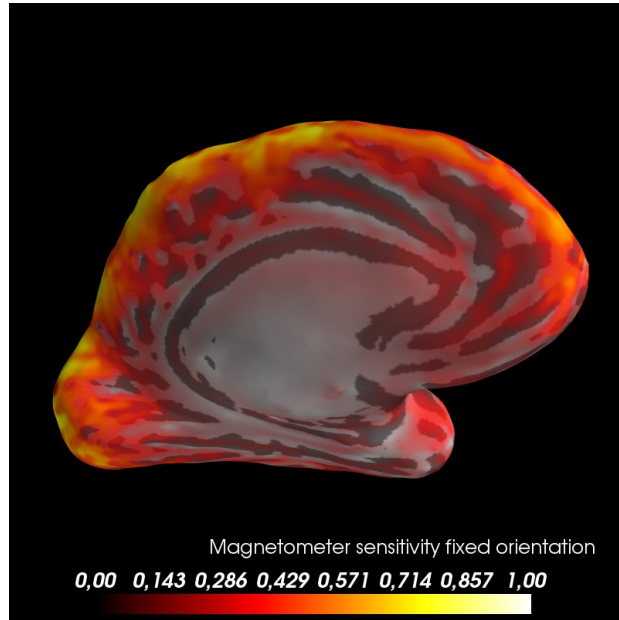


Sensitivity

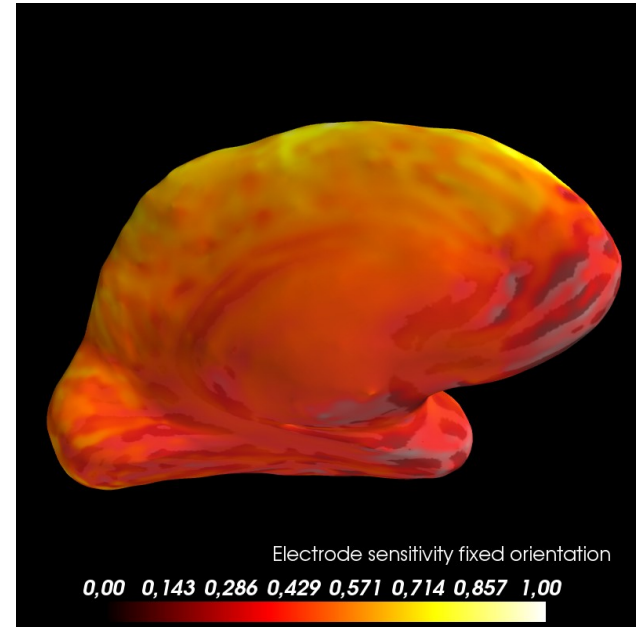


Sensitivity

MEG



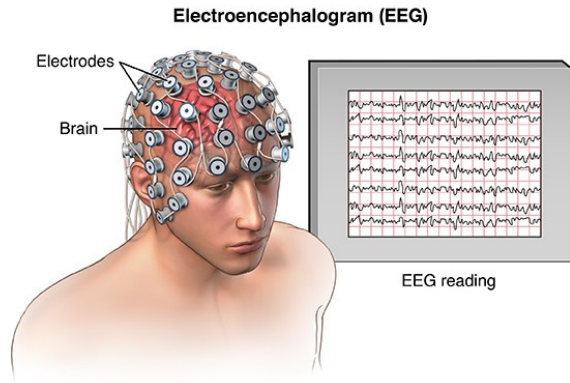
EEG



How to measure

Types of sensors

ELECTRODES

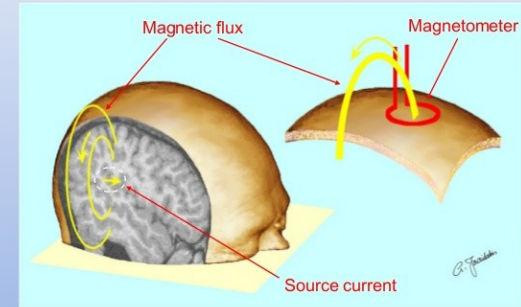


Don't have the details
for these images

Measures electric potential [V] (the volume currents)
(relative to a reference electrode)

MAGNETOMETERS

Magnetic flux from source currents

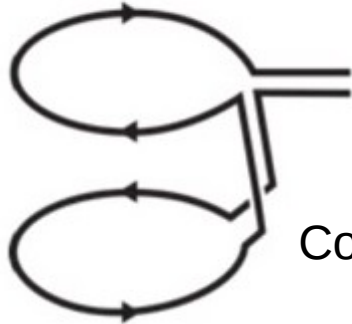


Measures magnetic flux [T]
(strength of magnetic field)

Types of sensors

AXIAL GRADIOMETERS

Axial gradiometer



PLANAR GRADIOMETERS

Planar gradiometer

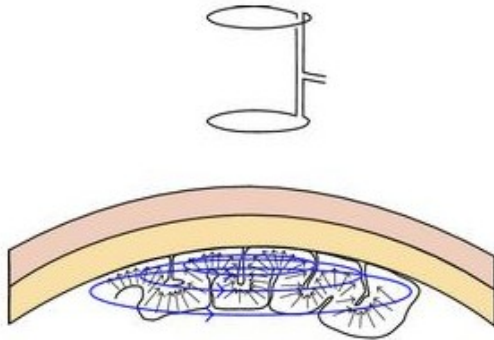


Consists of two magnetometers:

Measures magnetic gradient [T/m]
(how much strength of magnetic field
changes over two locations)

Types of sensors

AXIAL GRADIOMETERS

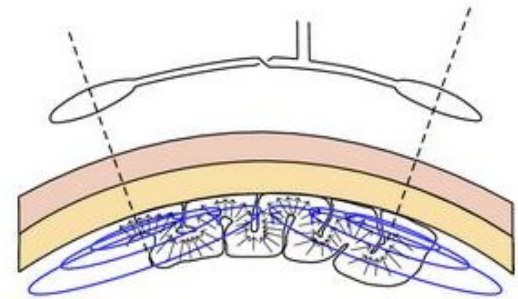


Axial gradiometer
(tangential)

Consists of two magnetometers:

Measures magnetic gradient [T/m]
(how much strength of magnetic field changes over two locations)

PLANAR GRADIOMETERS

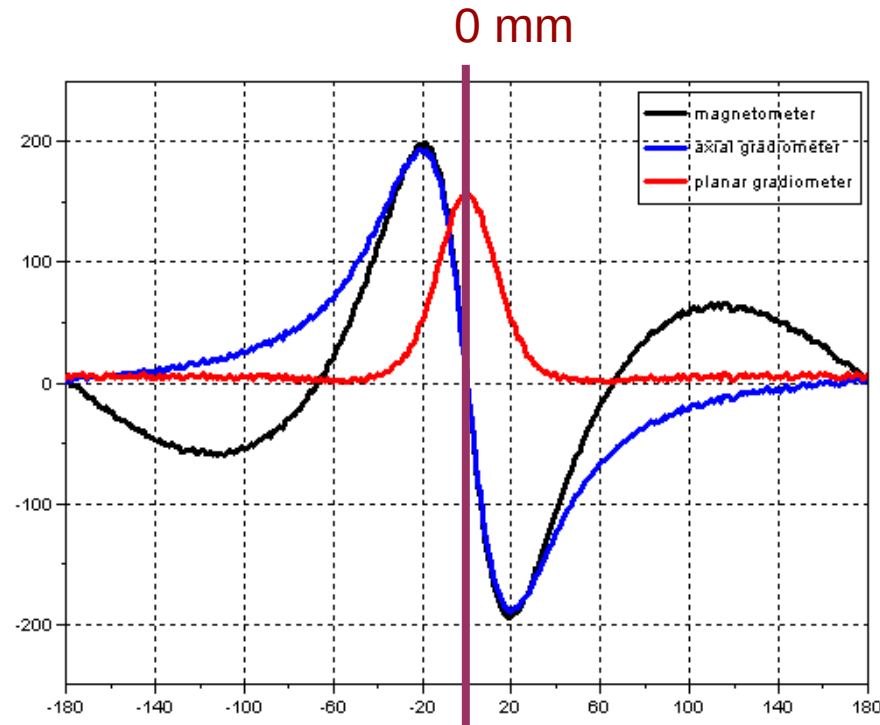


Planar gradiometer
(tangential)

Sensitivity profiles

sensitivity
to source

Courtesy of Luzia Troebinger,
University College London



distance to source in x or
y direction

MEGIN[®]



- 102 chips
 - each with a magnetometer
 - and two orthogonal planar gradiometers
 - thus 306 sensors total

CTF[®]



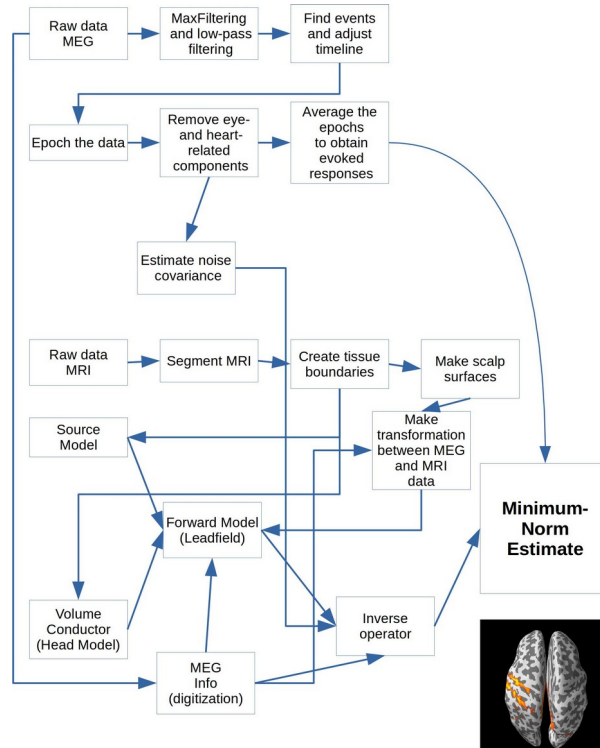
- 275 axial gradiometers
 - reference channels for noise cancellation

Interim summary

- The bulk of the MEG signal comes from post-synaptic potentials
 - In open field arrangements
- MEG is mainly sensitive to tangential sources
- We can use the equivalent current dipole to model the activity over a given patch (of cortex)
- In a volume conductor, currents will spread and the measured magnetic field will be (roughly) proportional to the primary current

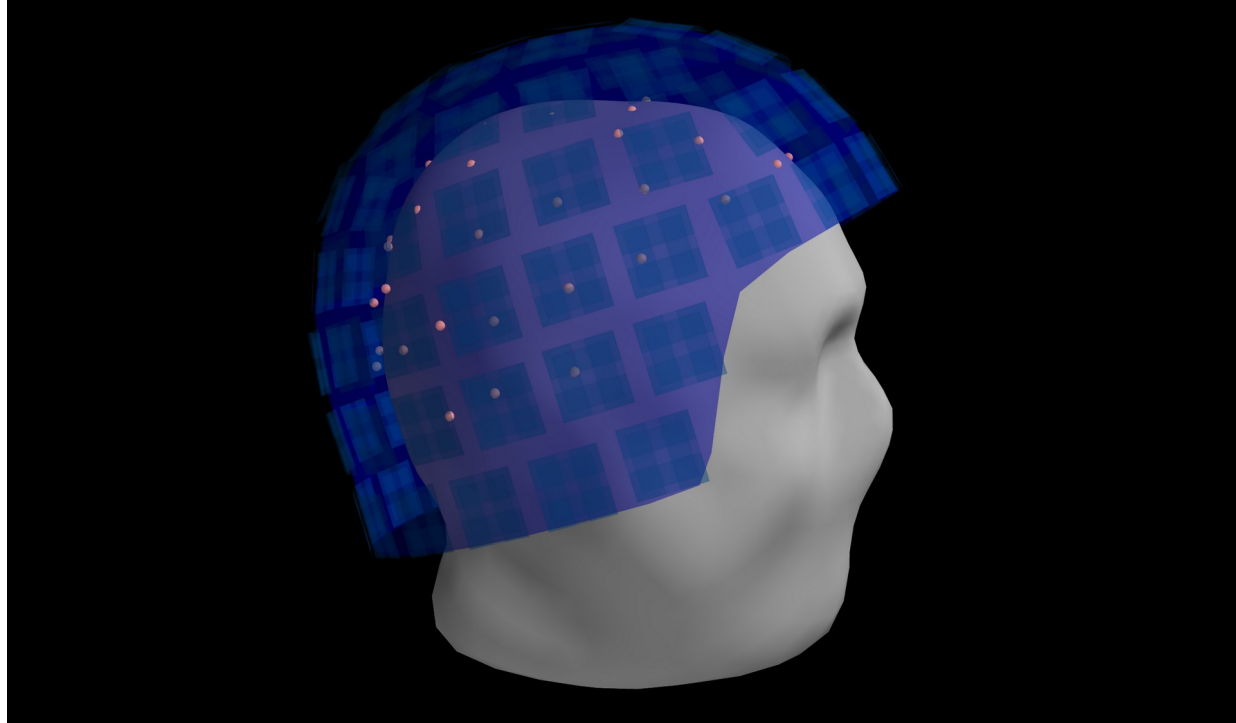
Evoked responses

An MEG pipeline

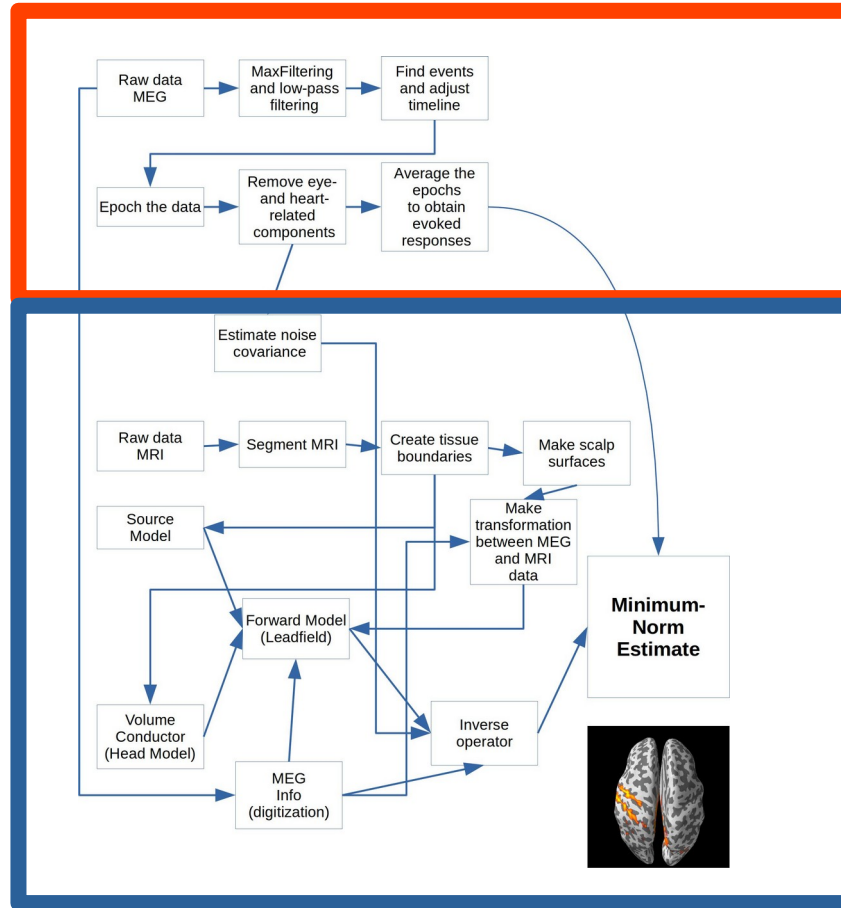


Andersen, 2018

Head and sensors



Sensor analysis



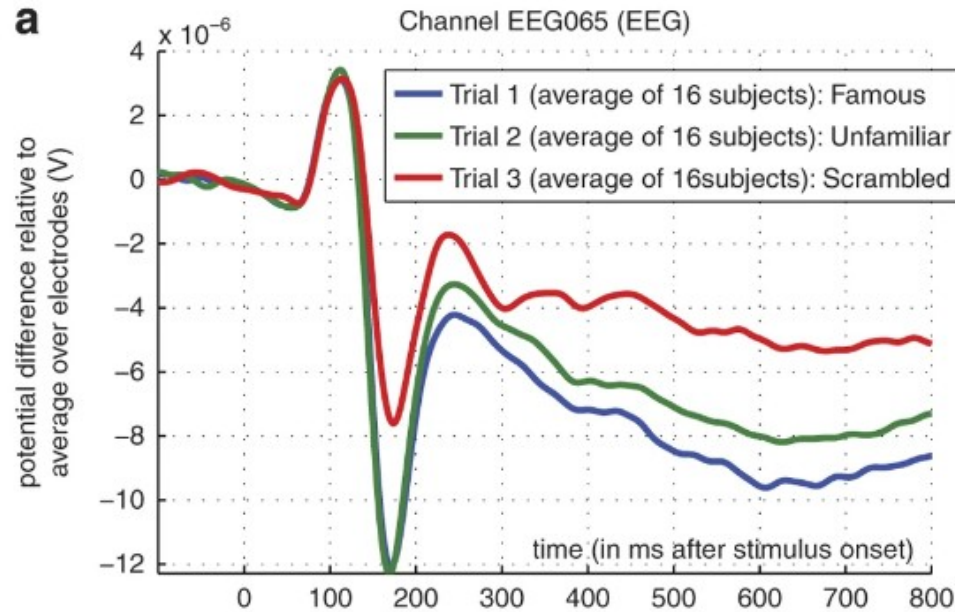
Andersen, 2018

Source analysis

Learning goals

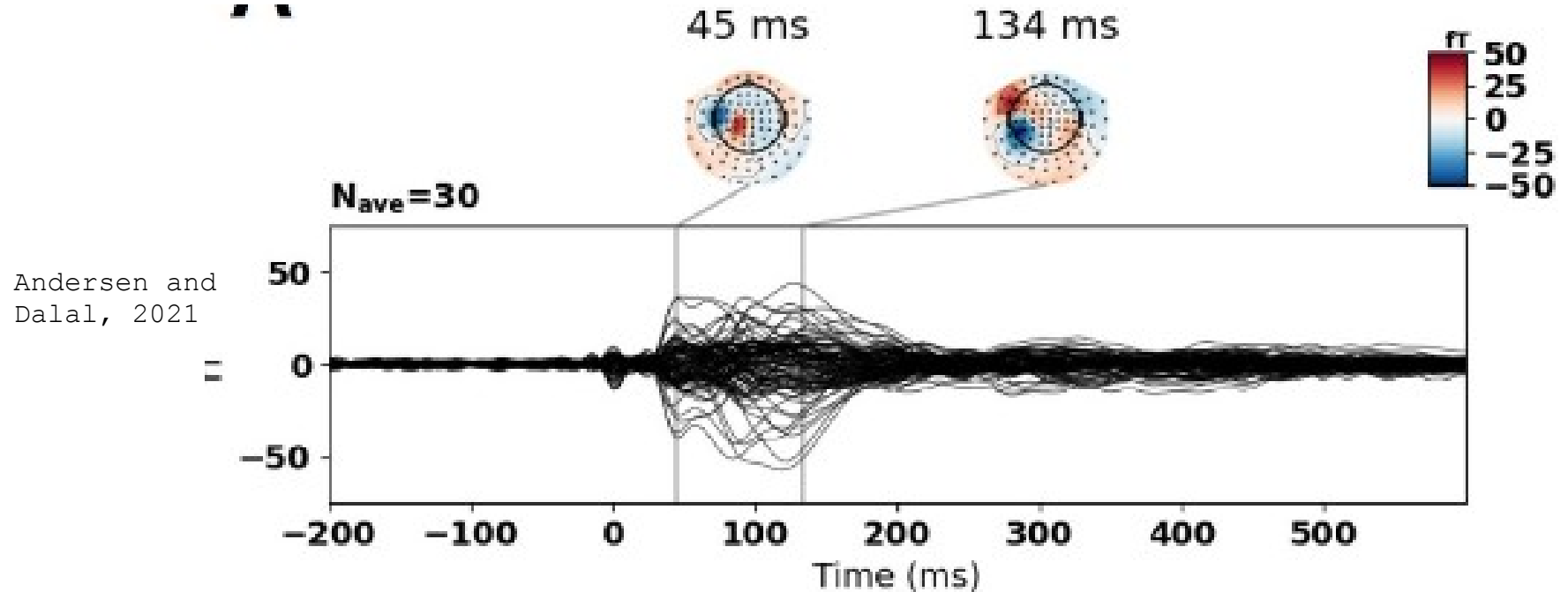
- Learning
 - the identification of physiological artefacts
 - strategies for increasing the signal-to-noise ratio
 - how to use epochs to create an evoked response
 - the importance of epochs being in-phase for an averaging strategy to be feasible
 - to do rough source localization on dipolar activations

Evoked responses – visual and face



Wakeman and
Henson, 2015

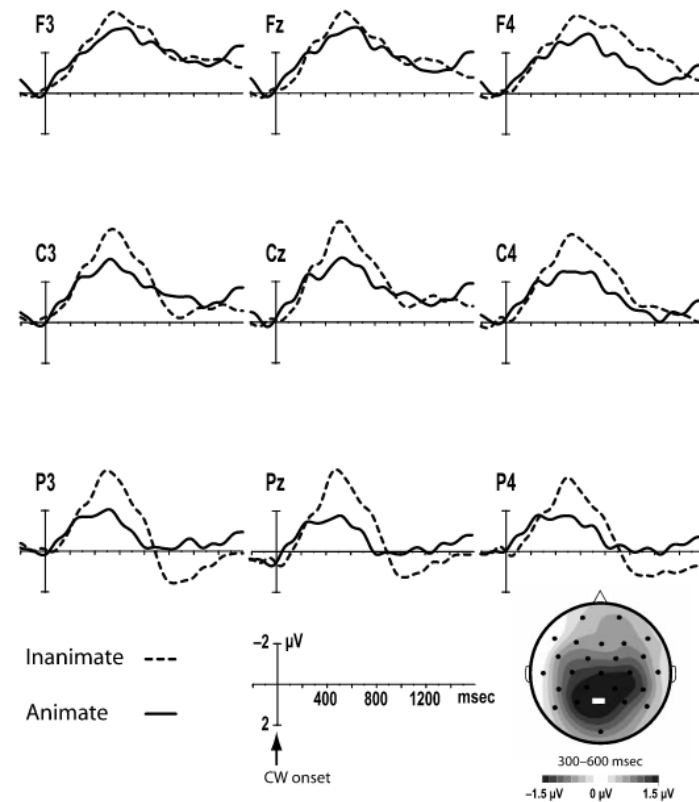
Evoked responses – tactile



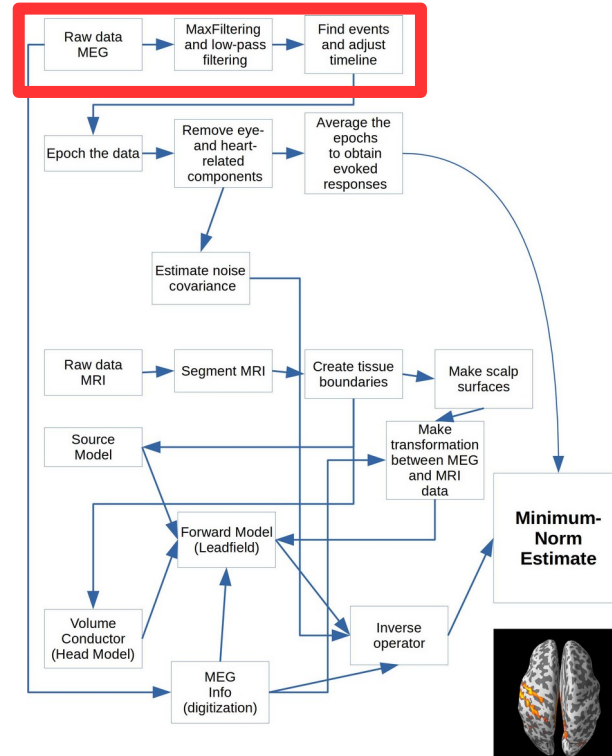
Evoked responses – linguistic surprise

Nieuwland and van
Berkum, 2006

Once upon a time a psychotherapist was consulted in her home office by a *yacht/sailor* with emotional problems. The yacht/sailor confided her that everything in life had gone wrong and started crying. The psychotherapist consoled the yacht/sailor by stating that everybody experiences these kinds of trouble every now and then. But the yacht/sailor doubted whether to continue outlining his problems to her. The psychotherapist advised the yacht/sailor to be honest not only with her, but especially with himself. At that moment the yacht/sailor cried out that he was absolutely terrified of water.

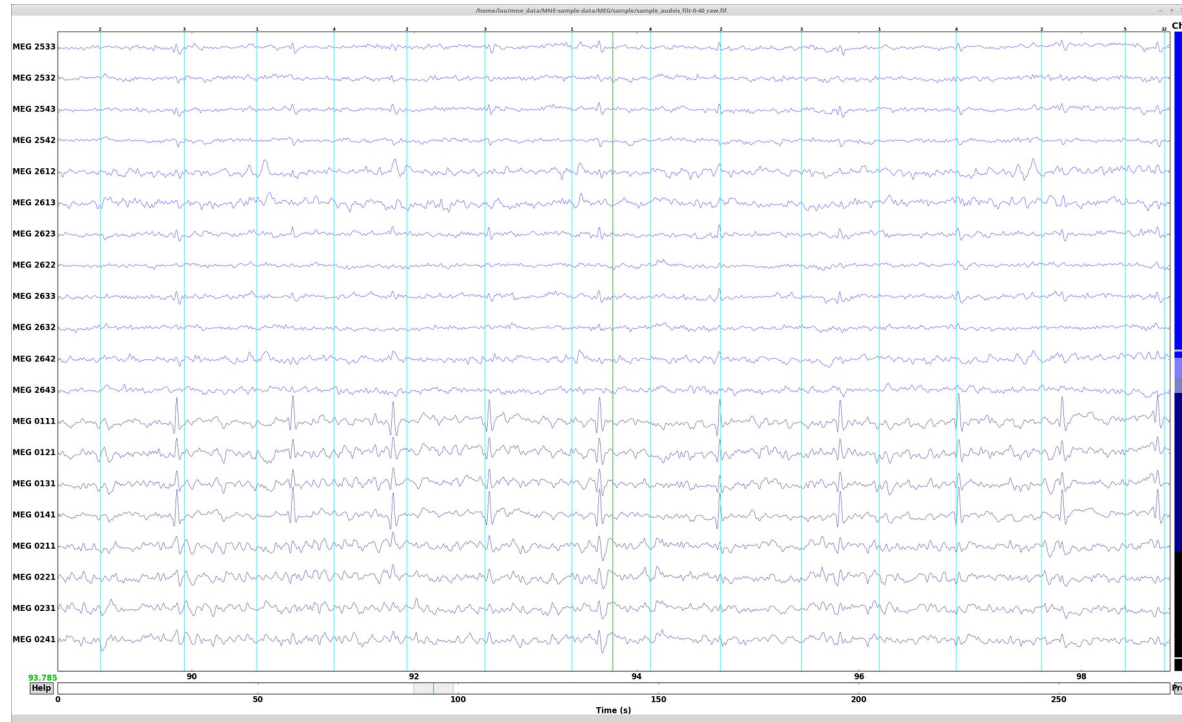


An MEG pipeline

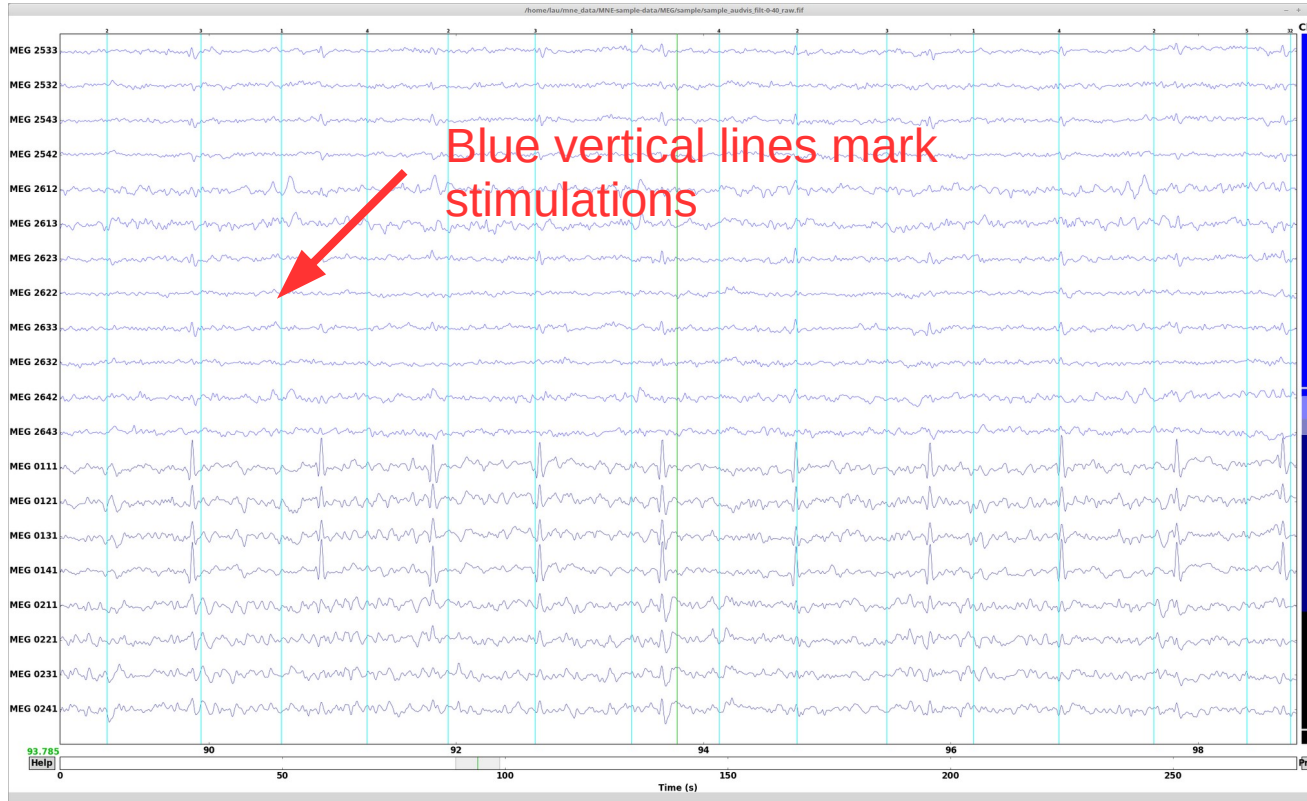


Andersen, 2018

The raw signal (MEG)



The raw signal (MEG)?



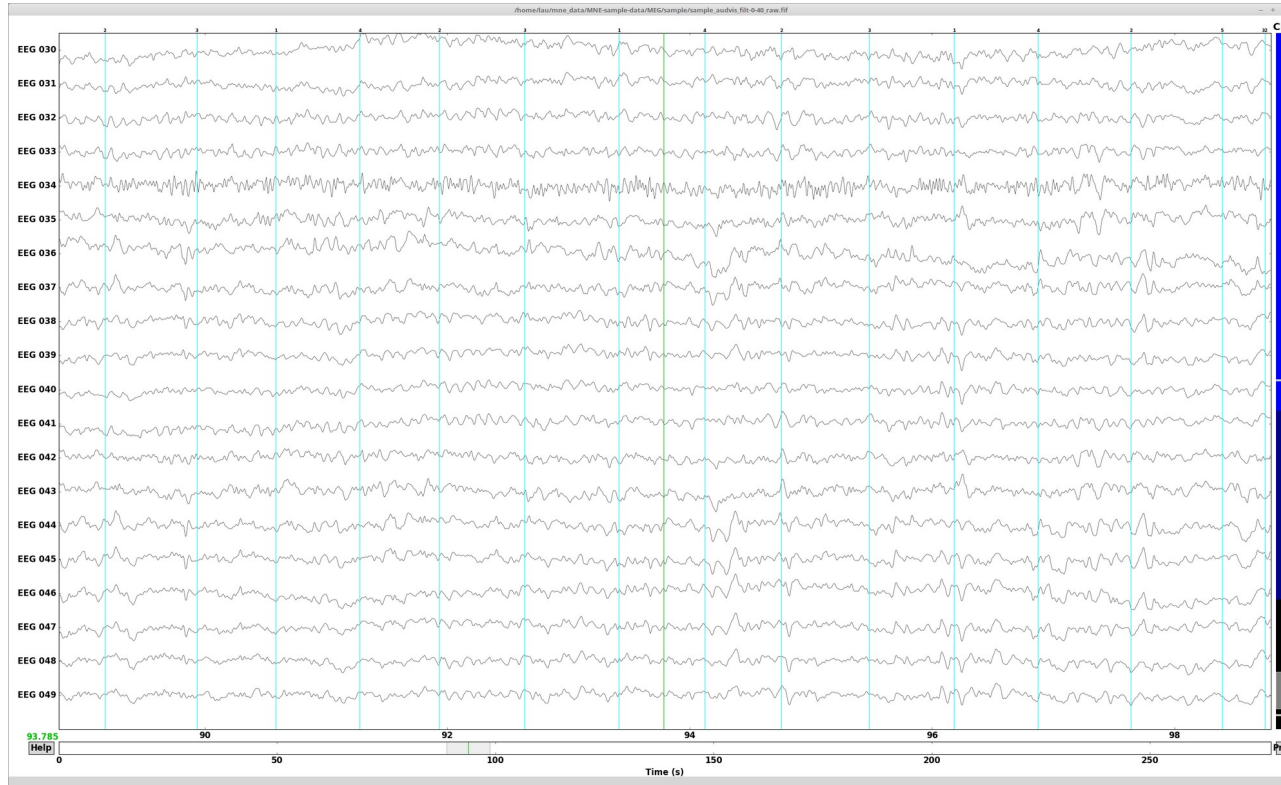
The raw signal (MEG)?



The raw signal (MEG)?



The raw signal (EEG)



Signal-to-Noise Ratio is not sufficiently high to see the brain responses to the auditory stimulation in the raw signal

(This actually holds for most brain responses)

Signal-to-noise ratio (SNR)

Crude definition:

Signal: activity of interest

Noise: activity *not* of interest

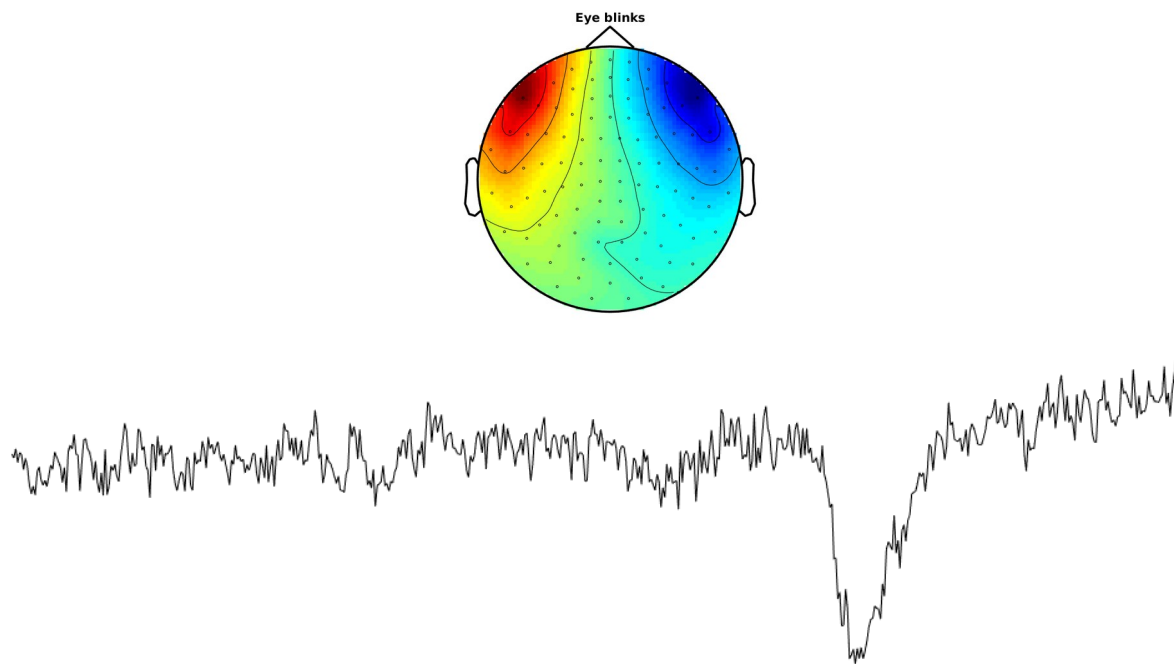
Formal Definition:

$$SNR = \frac{\text{Magnitude of signal}}{\text{Variance of noise}}$$

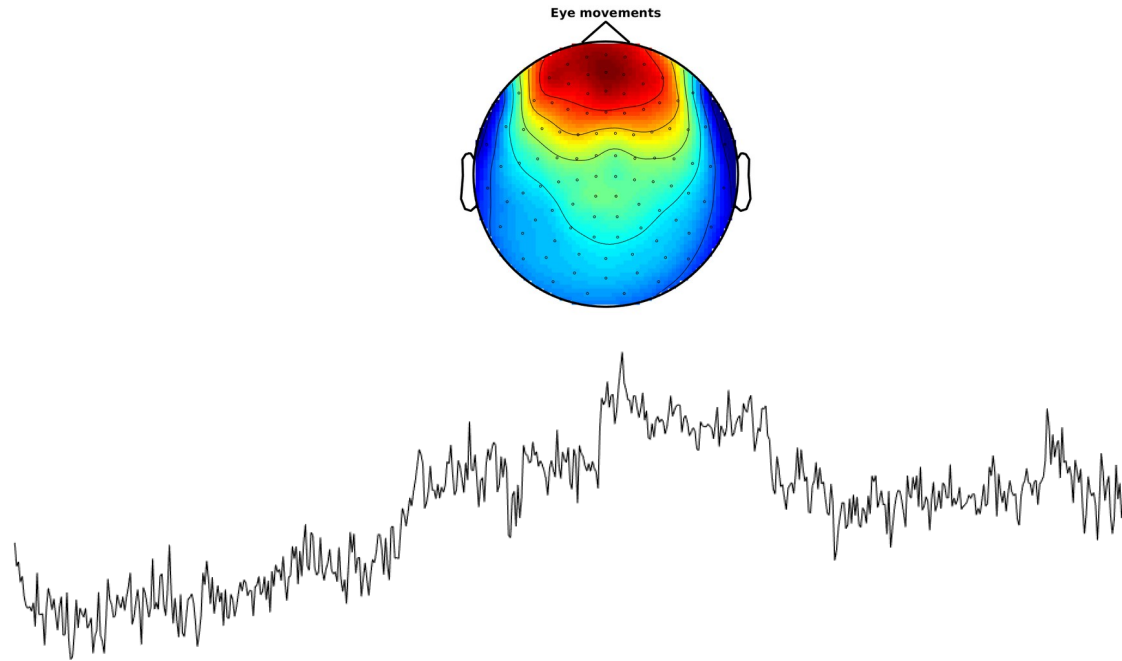
Examples of noise

- Eye blinks
- Eye movements
- Heart beats
- System noise
- External noise

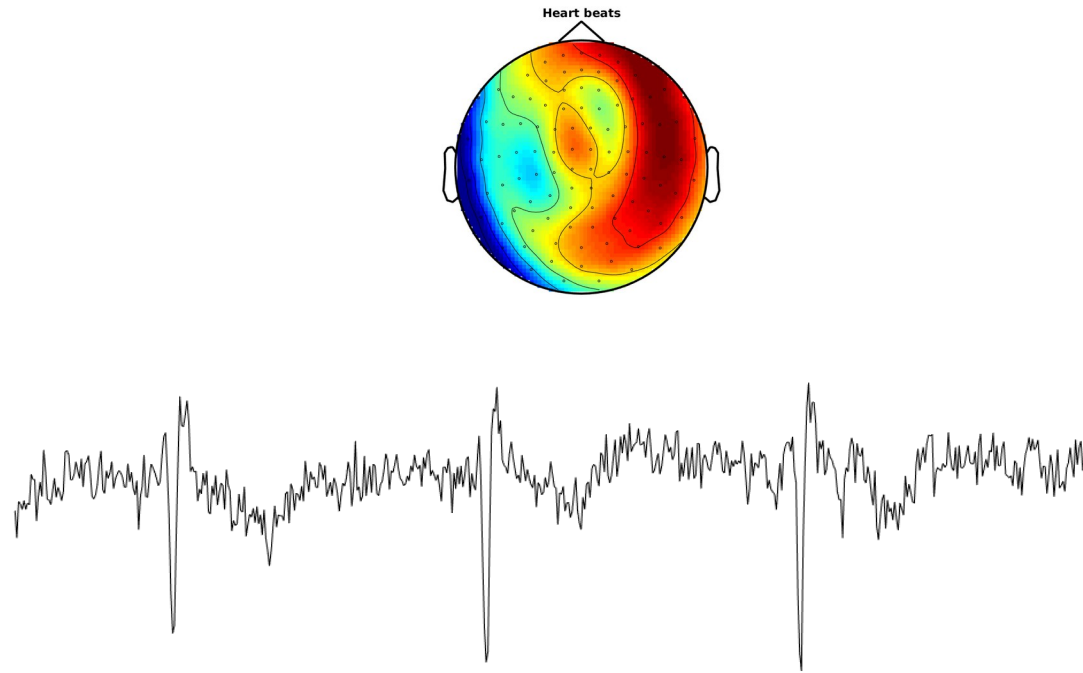
Eye blinks



Eye movements



Heart beats



How do we increase the
signal amidst all the
noise?

Increasing SNR

1) Filtering

- Remove contributions from frequencies not of interest

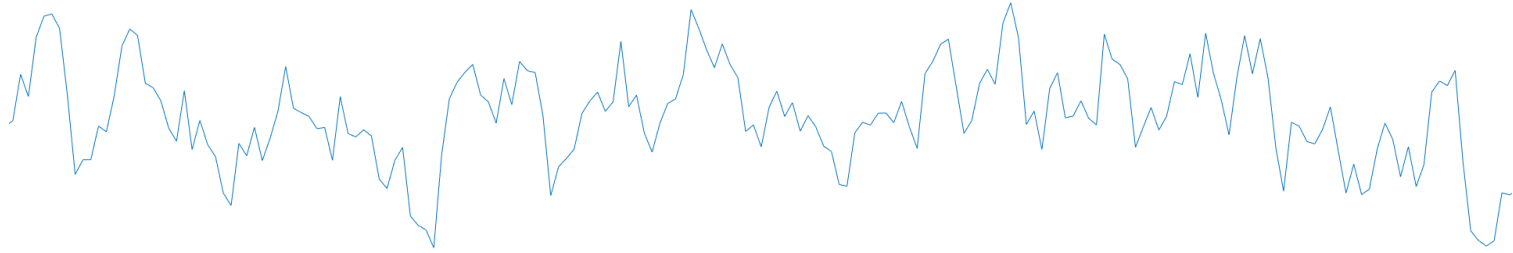
2) Clean data

- Reject trials with noise in them
- Remove activity associated with physiological noise

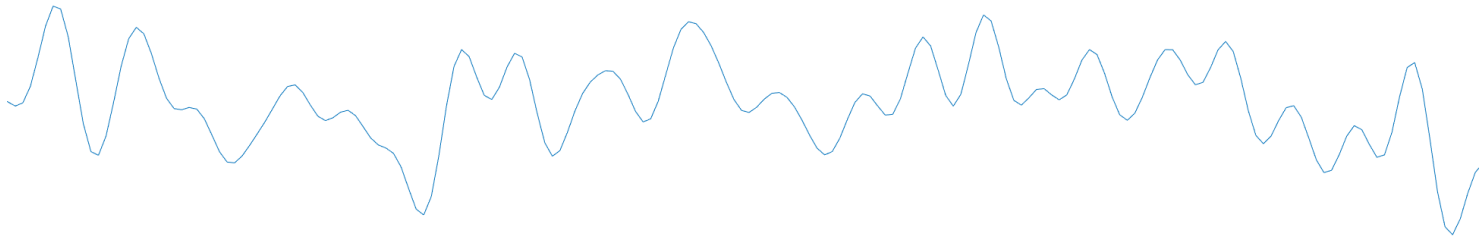
3) Average trials

- SNR increases
 - NB! The averaging strategy only works if responses are phase-locked

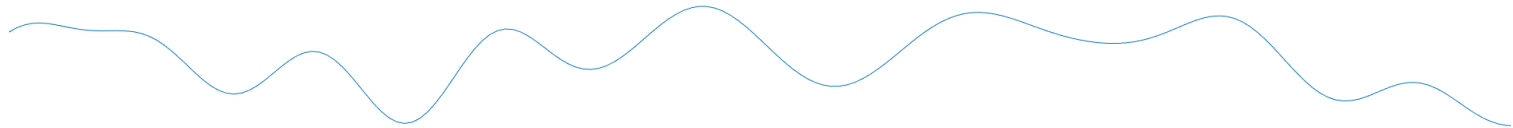
NO FILTERING



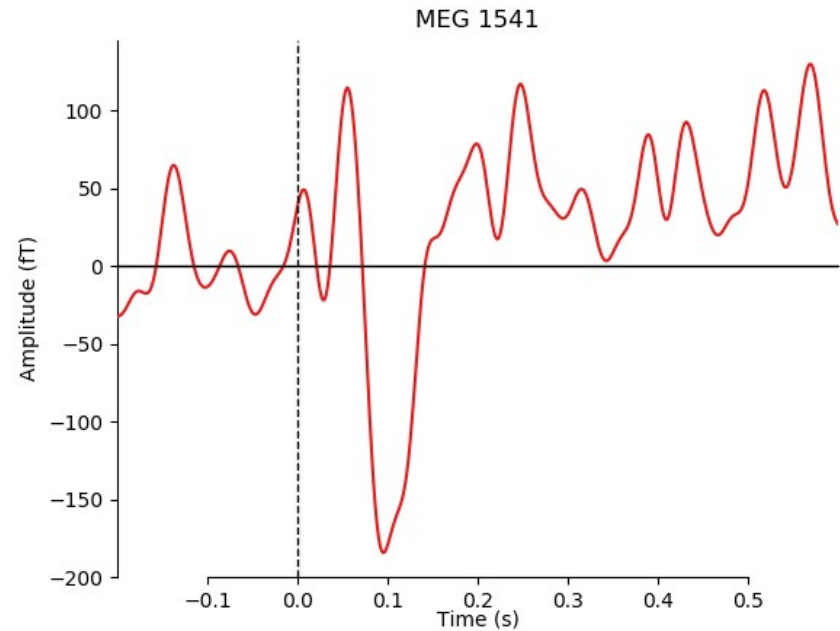
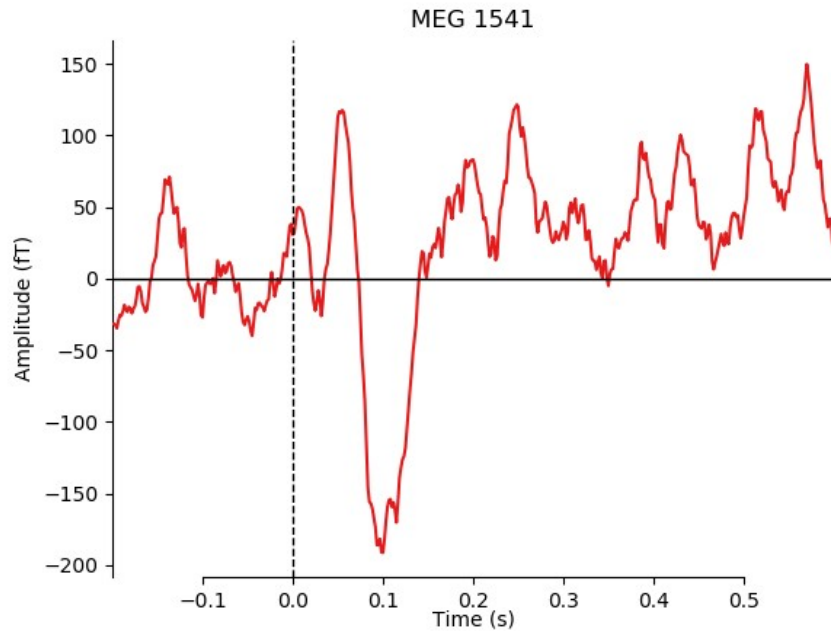
LOW-PASS 30 Hz (remove > 30 Hz)



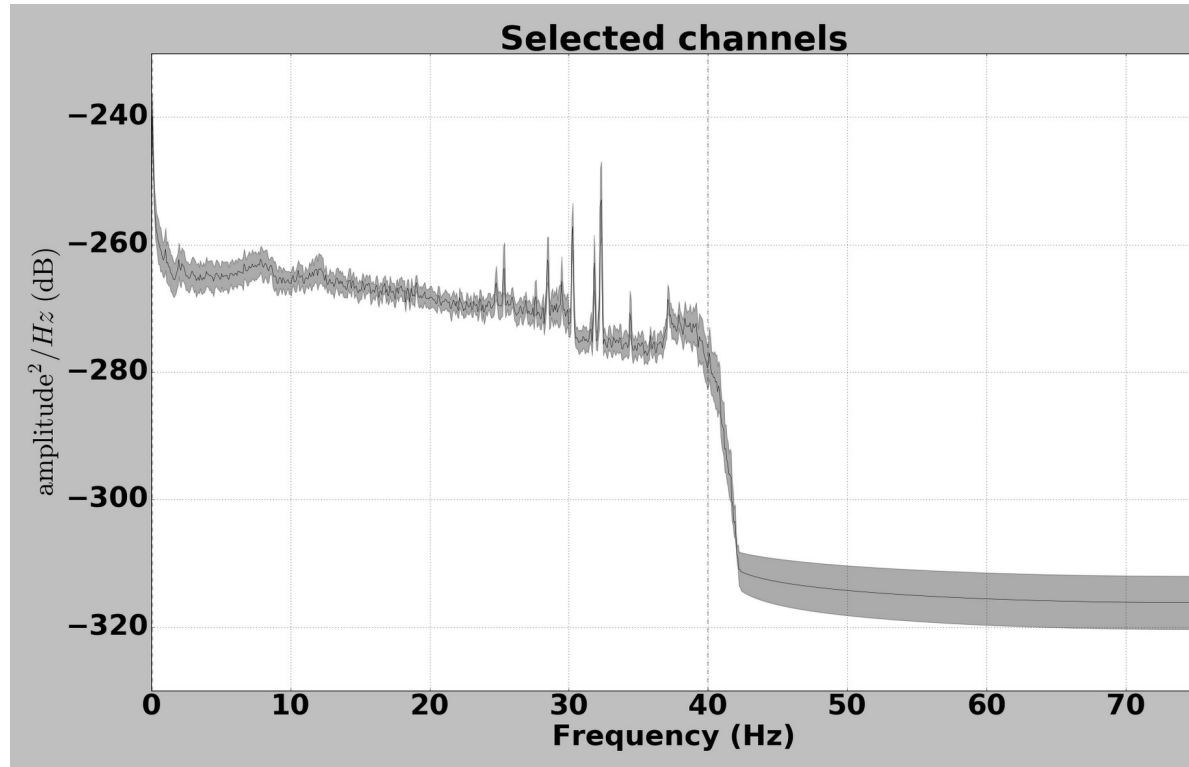
LOW-PASS 10 Hz (probably also removing brain signal)



With or without 30 Hz lowpass filter



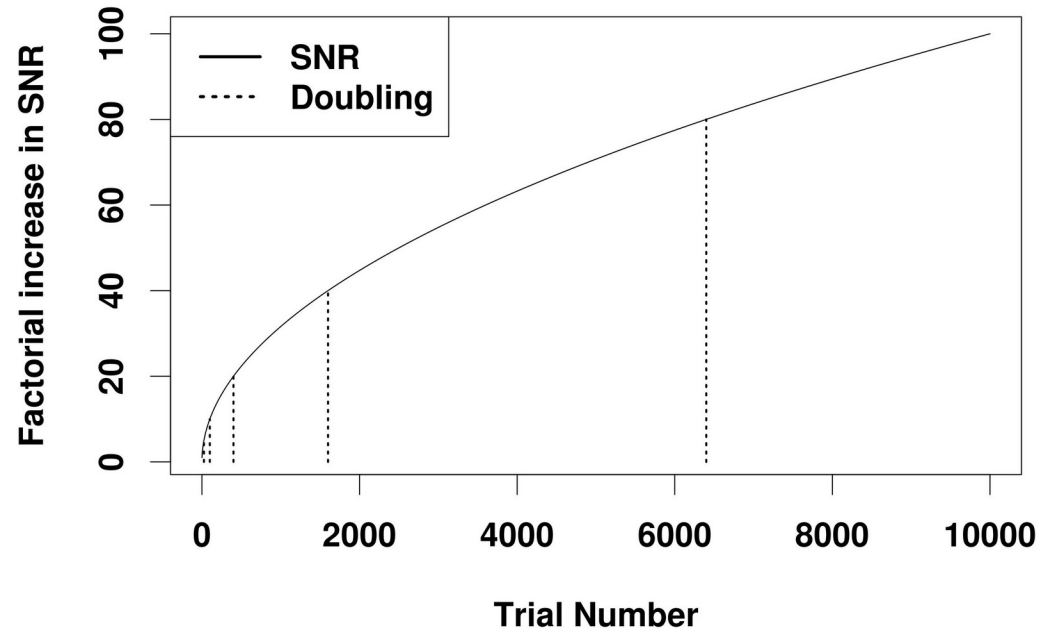
Another look at filtering



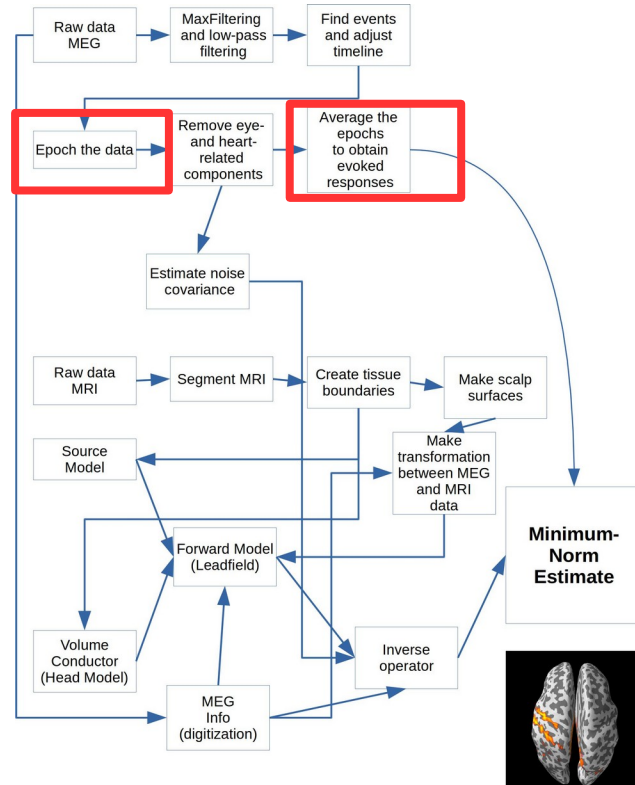
Averaging

DIMINISHING RETURNS

Relation between number of trials and SNR

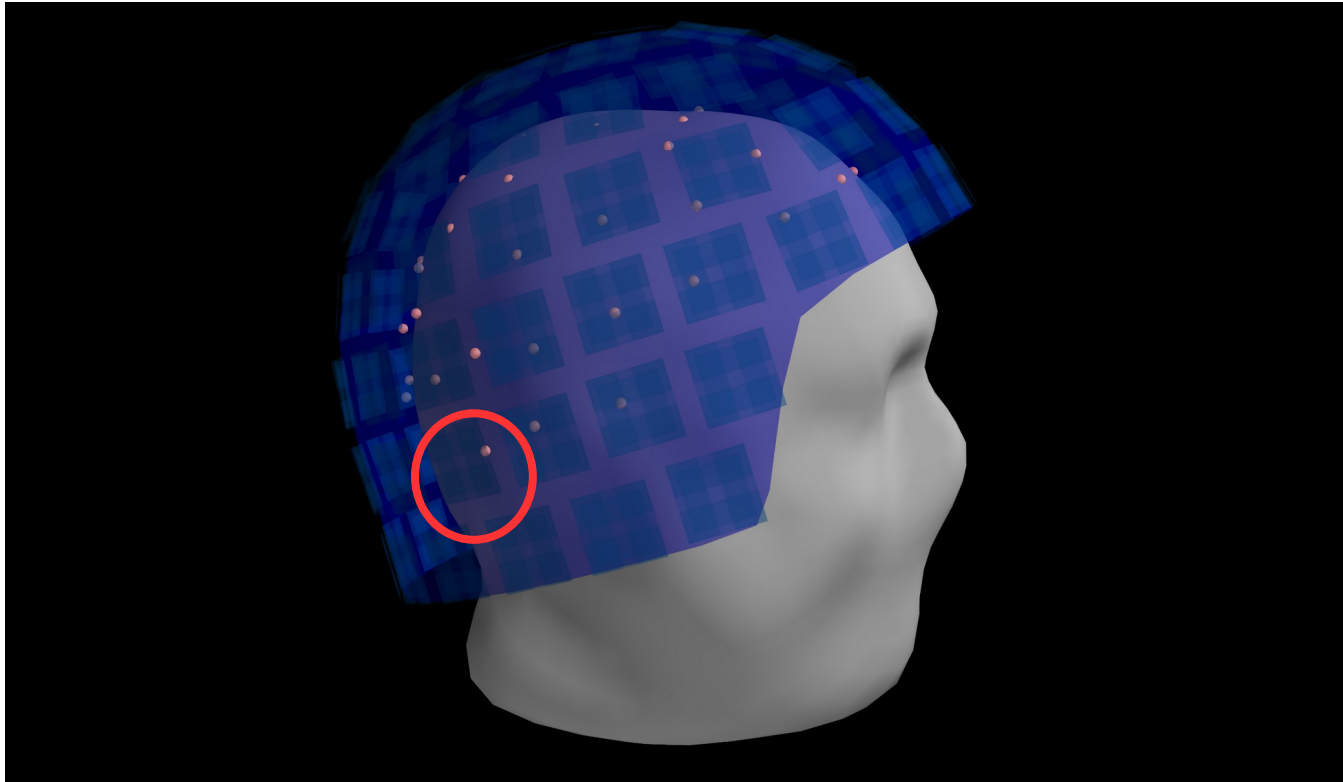


An MEG pipeline

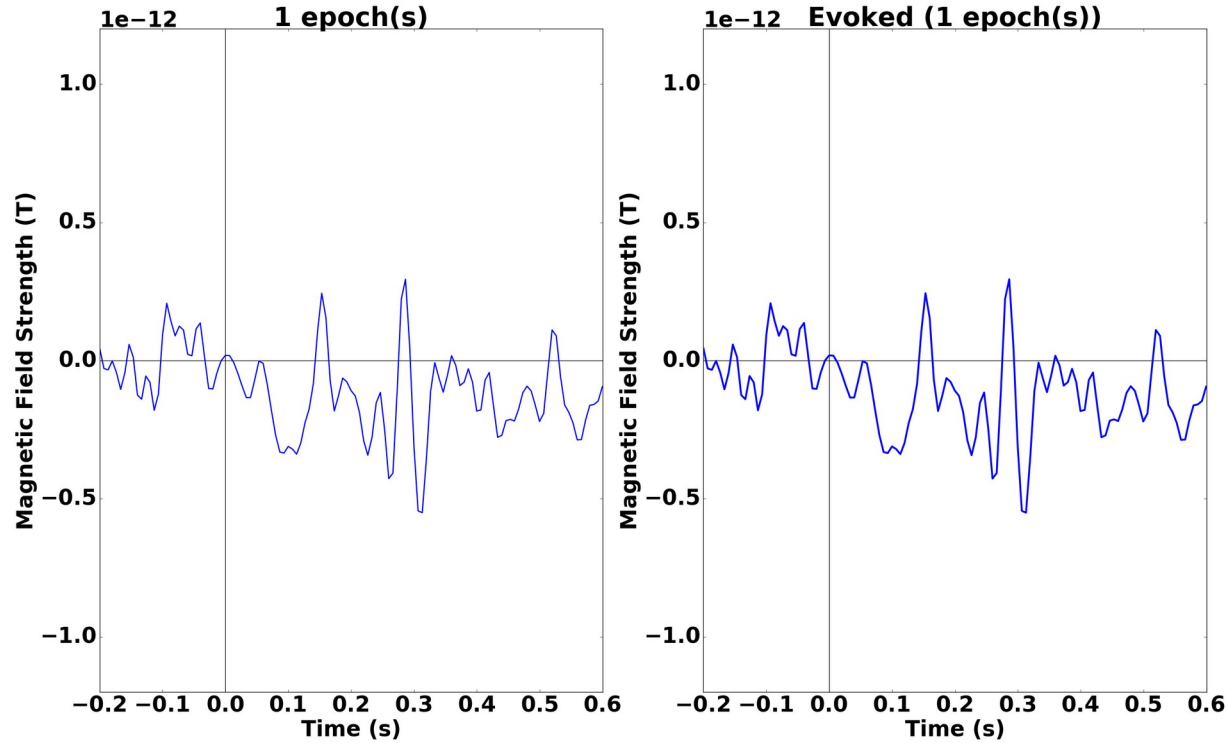


Andersen, 2018

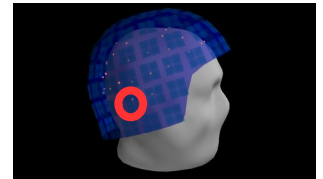
Head and sensors



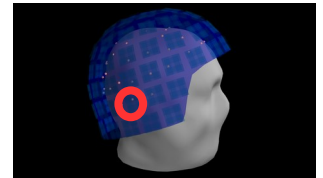
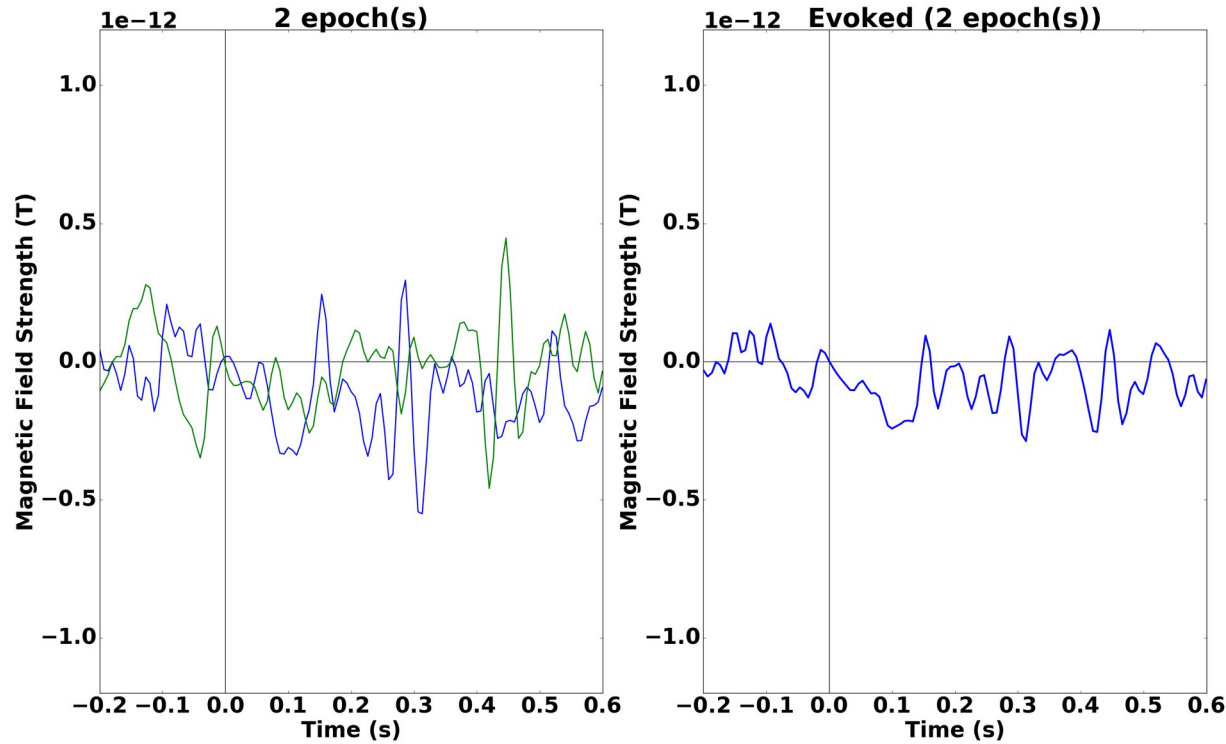
Epochs and Evoked



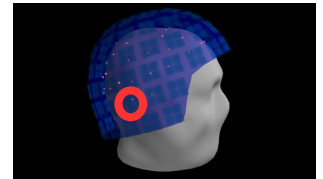
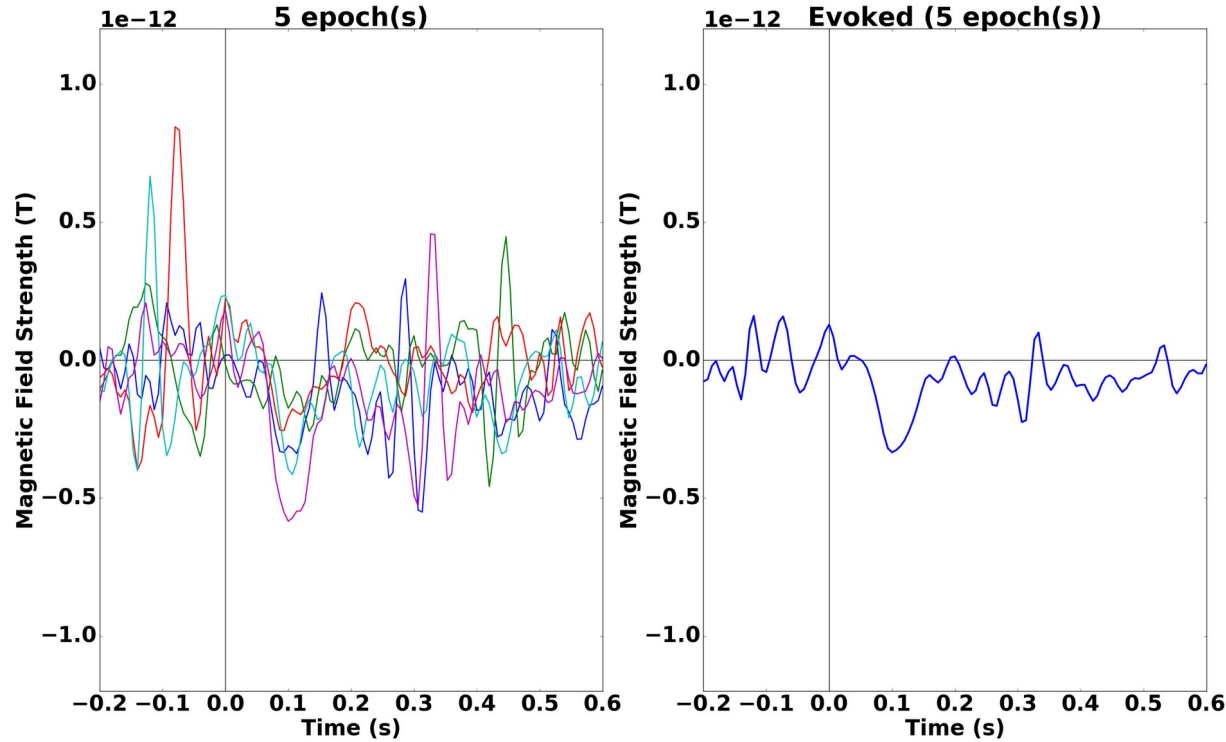
CC BY Licence 4.0: Lau Møller Andersen 2026



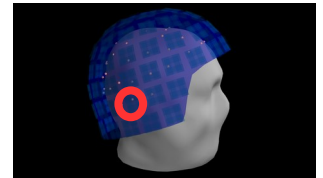
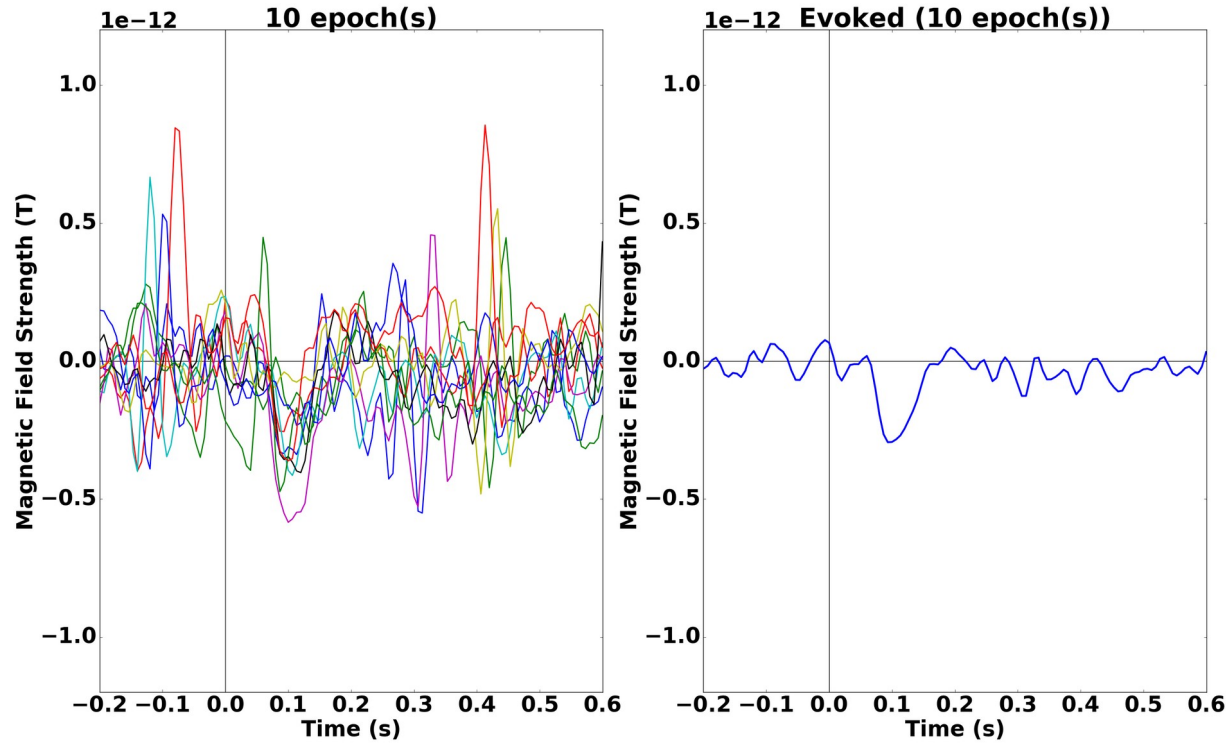
Epochs and Evoked



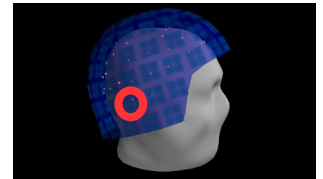
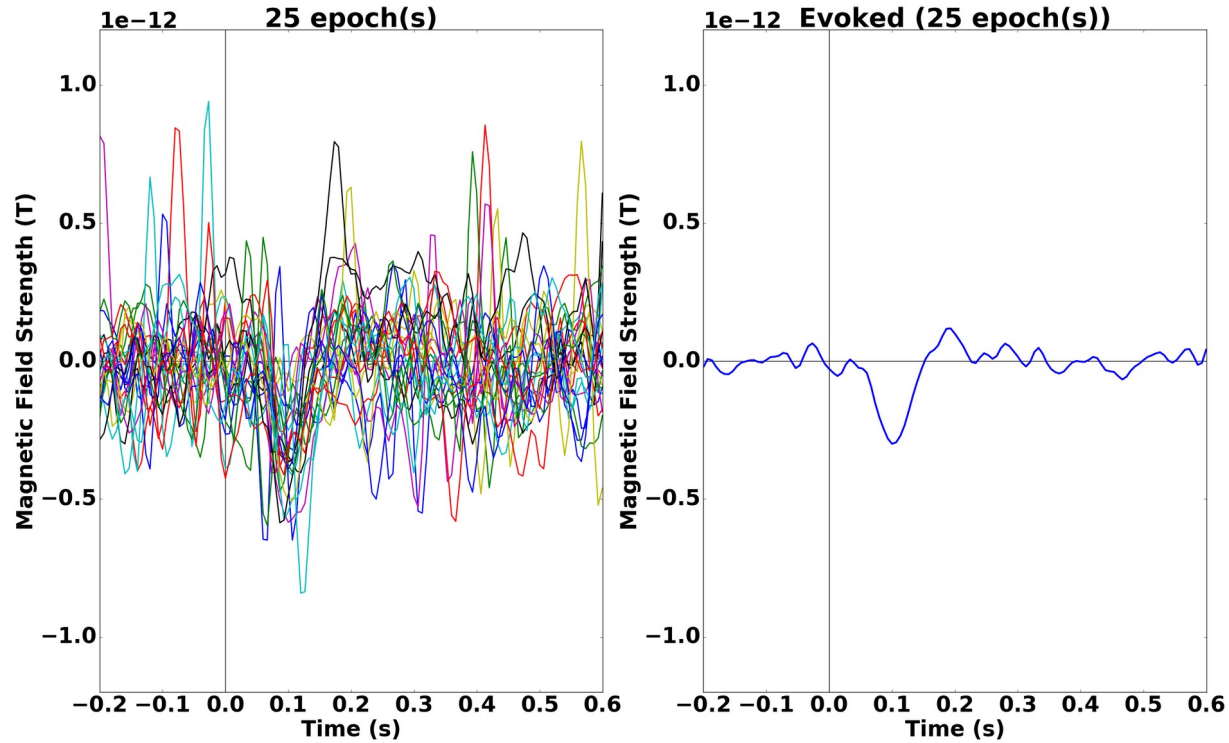
Epochs and Evoked



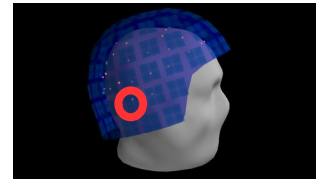
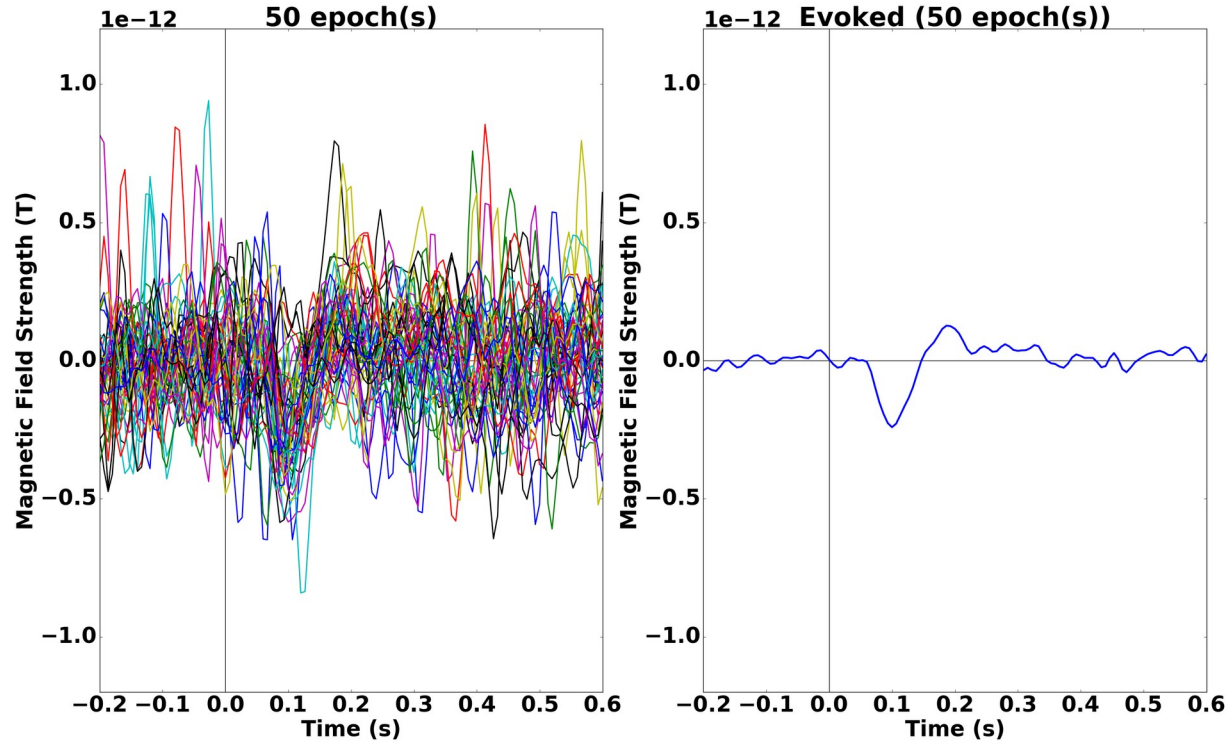
Epochs and Evoked



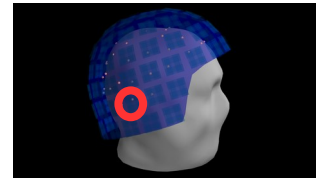
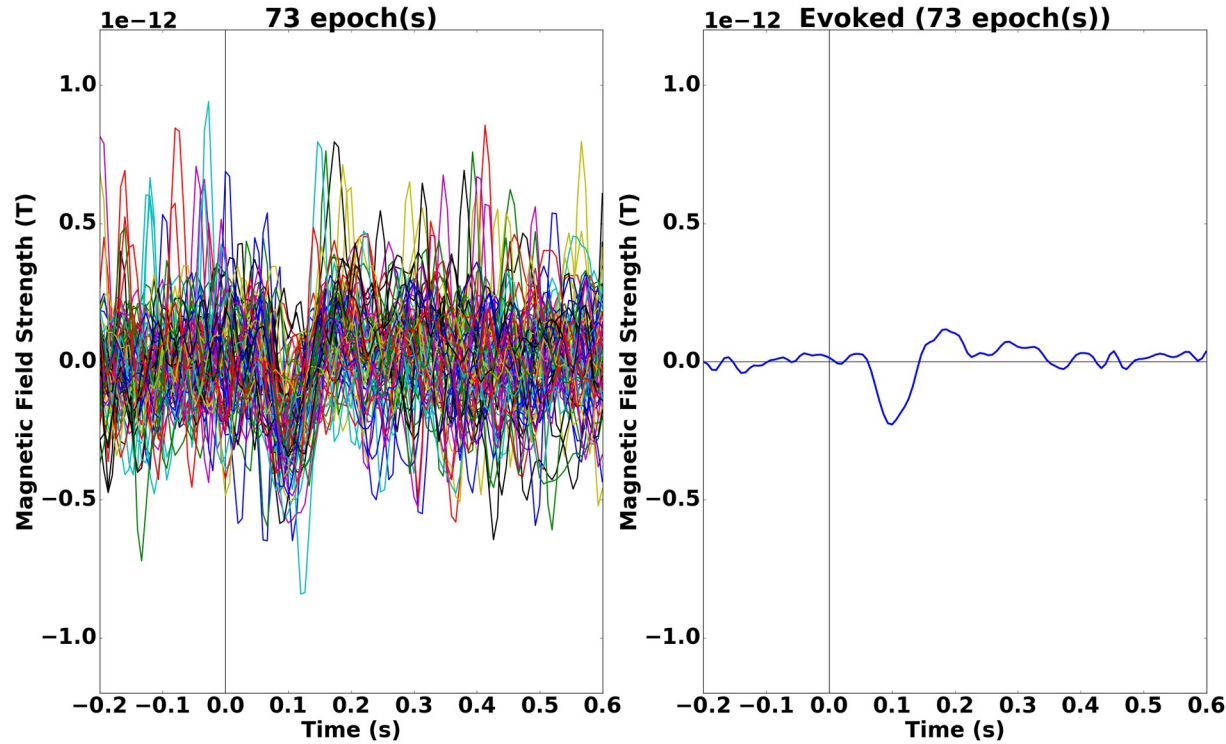
Epochs and Evoked



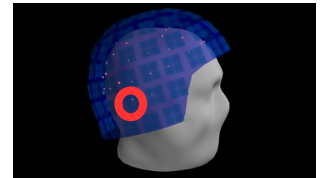
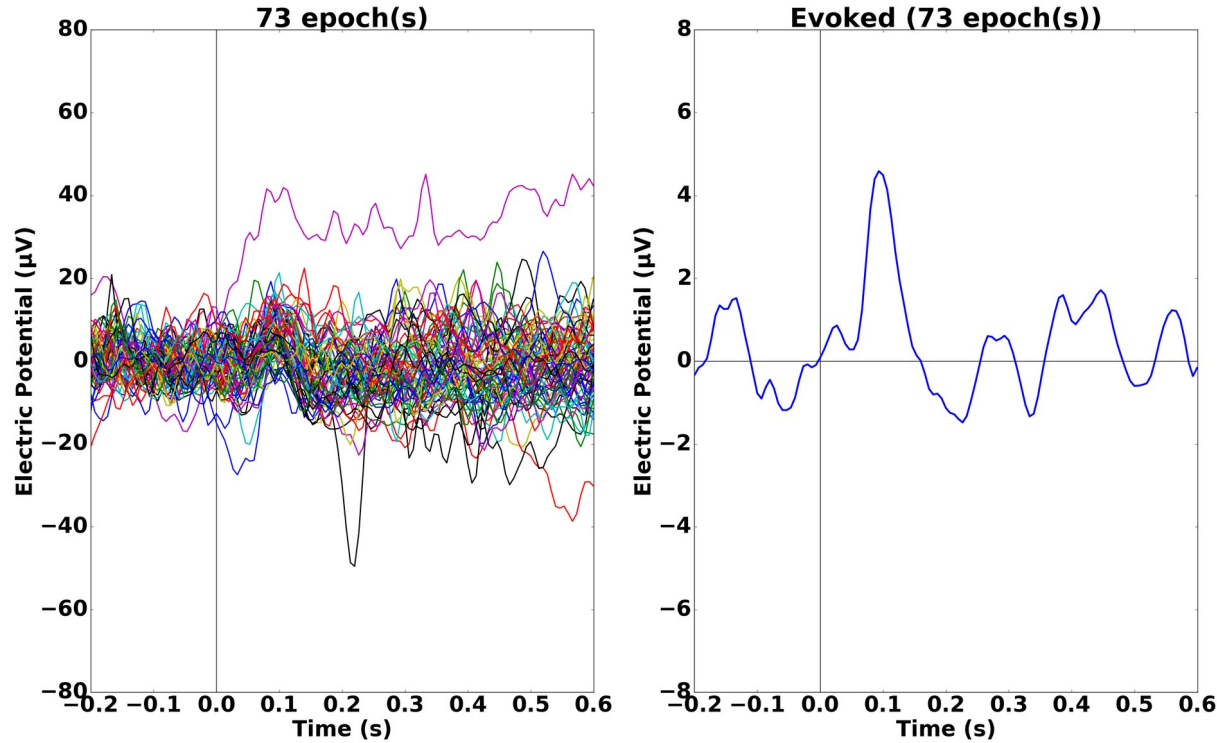
Epochs and Evoked



Epochs and Evoked



The same with EEG



```
### FIND EVENTS
```

```
events_sample = mne.find_events(raw_sample)  
mne.viz.plot_events(events_sample, sfreq=raw_sample.info['sfreq']);
```

```
### EPOCH THE DATA
```

```
event_id = dict(LA=1) ## we'll just look at this one event
```

```
tmin = -0.200 # s
```

```
tmax = 0.600 # s
```

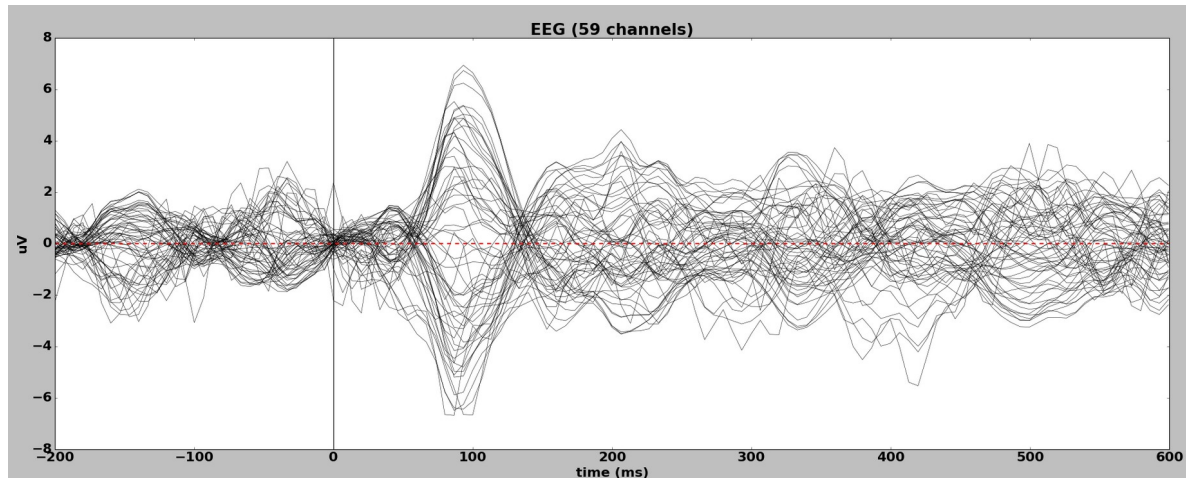
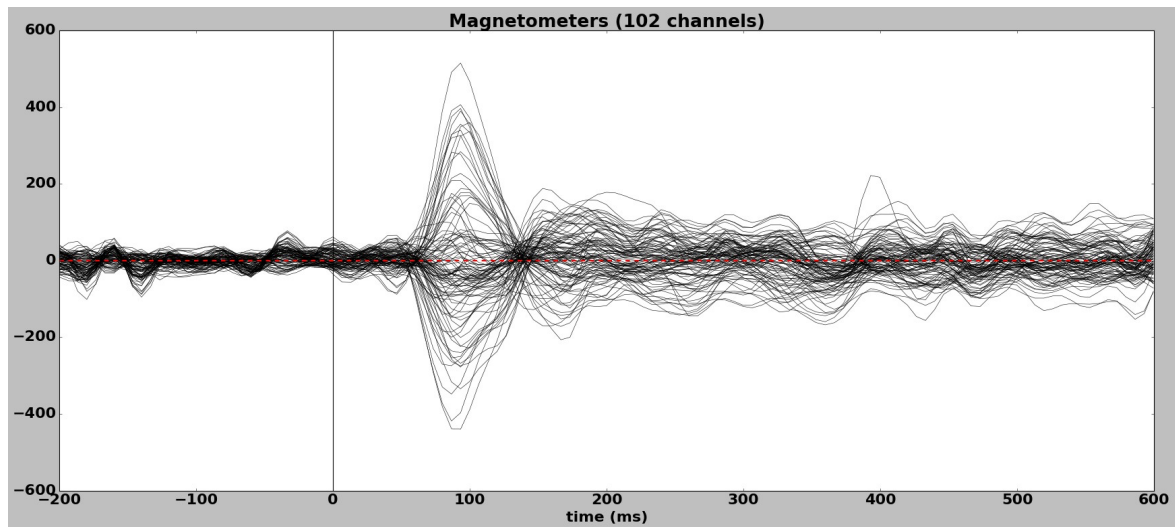
```
baseline = (None, 0) # s (from beginning to 0); for demeaning
```

```
## segment the data
```

```
epochs_sample = mne.Epochs(raw_sample, events_sample,  
                             event_id, tmin, tmax, baseline, preload=True)
```

```
### AVERAGE (EVOKED)
```

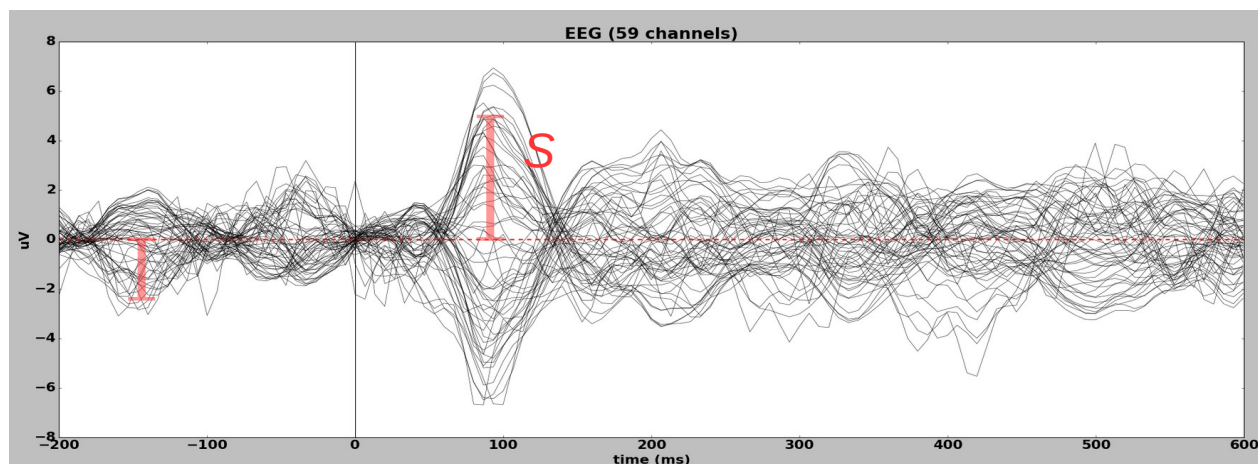
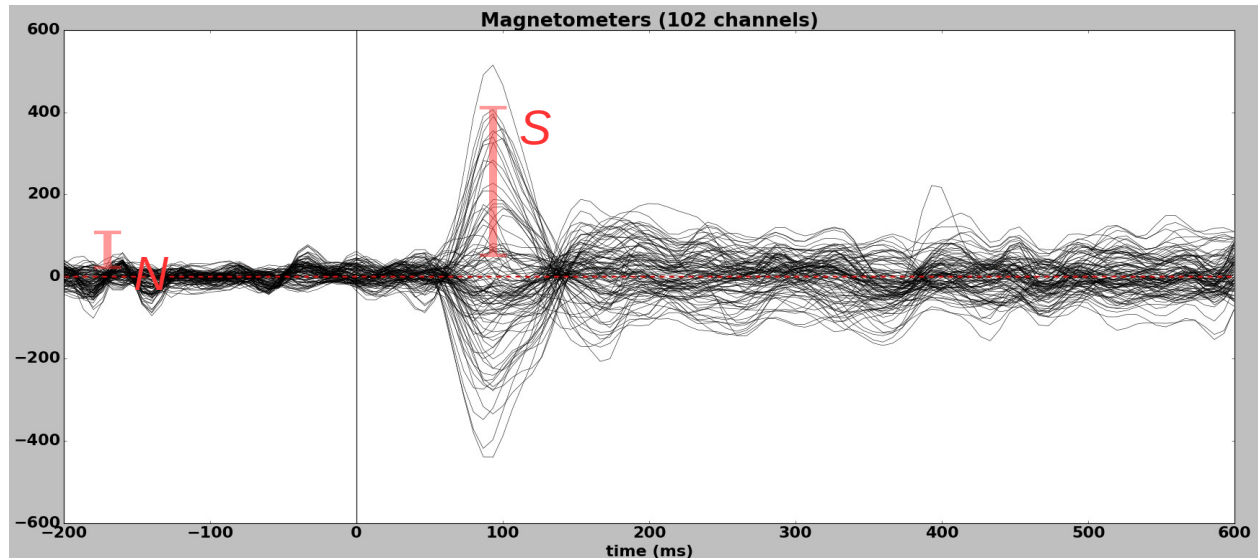
```
evoked_sample = epochs_sample.average()
```



The traces look very similar, but the signal-to-noise ratio is higher in the MEG than in the EEG

$$\text{SNR}_{\text{MEG}} = \frac{S}{N} = \frac{I}{I} \approx 4$$

$$\text{SNR}_{\text{EEG}} = \frac{S}{N} = \frac{I}{I} \approx 2$$



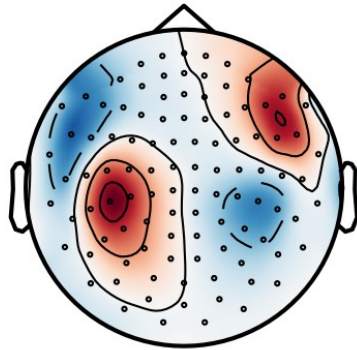
NOTE: These SNR estimates are for this particular experiment only. They are not necessarily generalizable

Topographies look very different

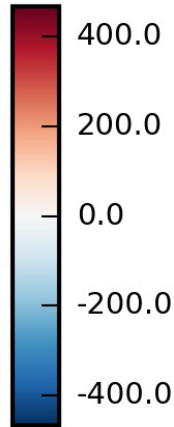
auditory_right mag

auditory_right eeg

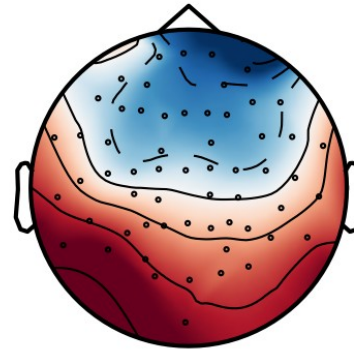
100 ms



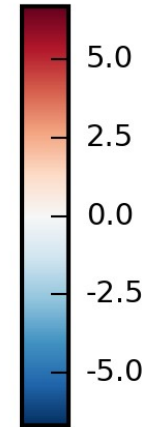
fT



100 ms



uV

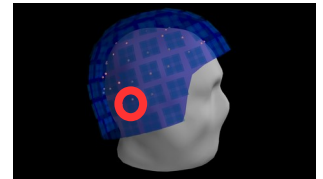
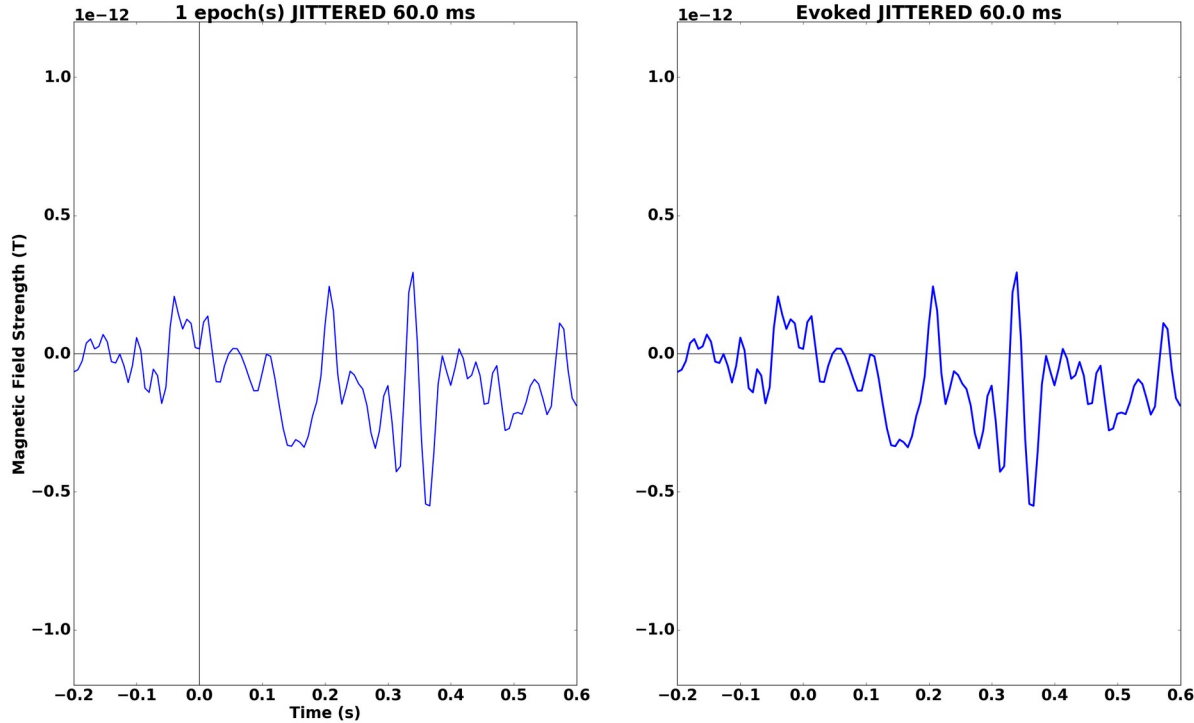


.au 1

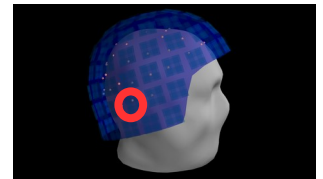
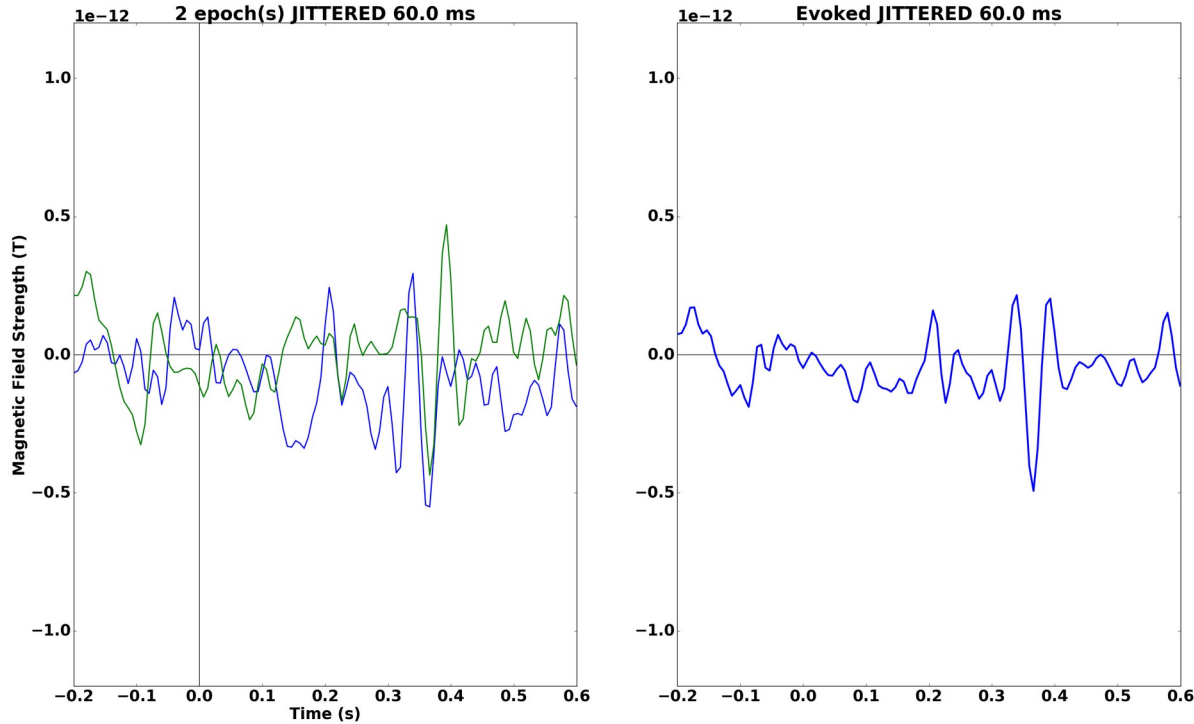
73 trials

FURTHER NOTE: Averaging only works as a strategy for optimizing SNR if responses are “phase-locked”, i.e. the time course is evolving in the same manner on all (most) trials

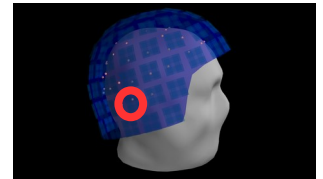
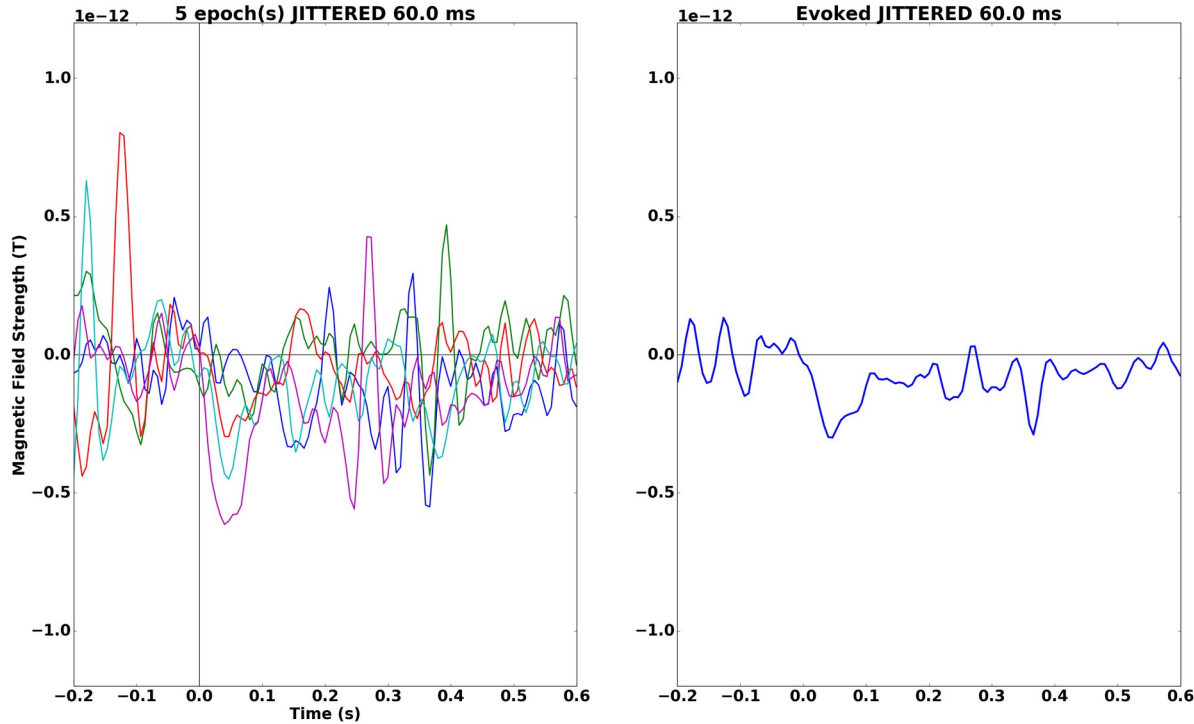
Epochs and Evoked JITTERED 60 ms



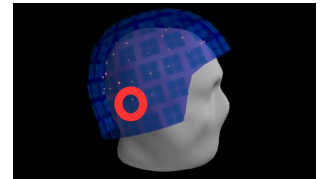
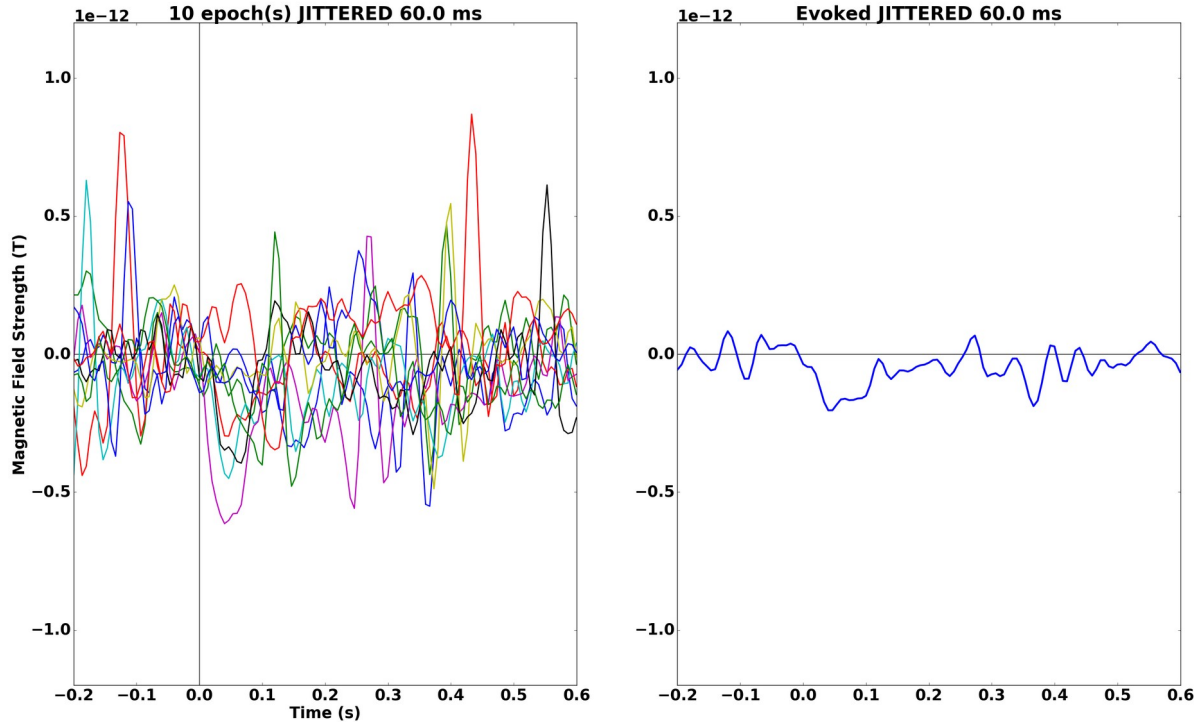
Epochs and Evoked JITTERED 60 ms



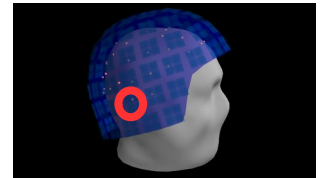
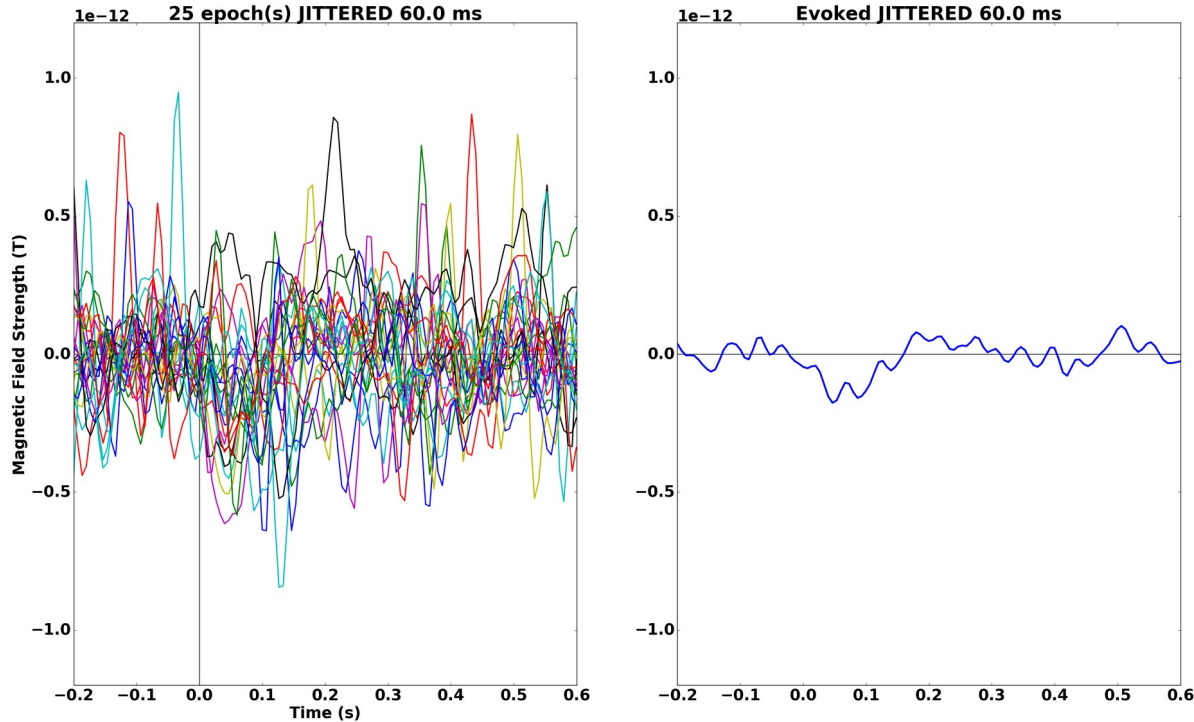
Epochs and Evoked JITTERED 60 ms



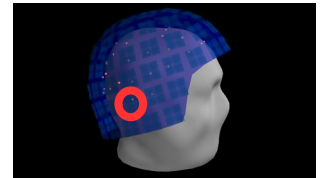
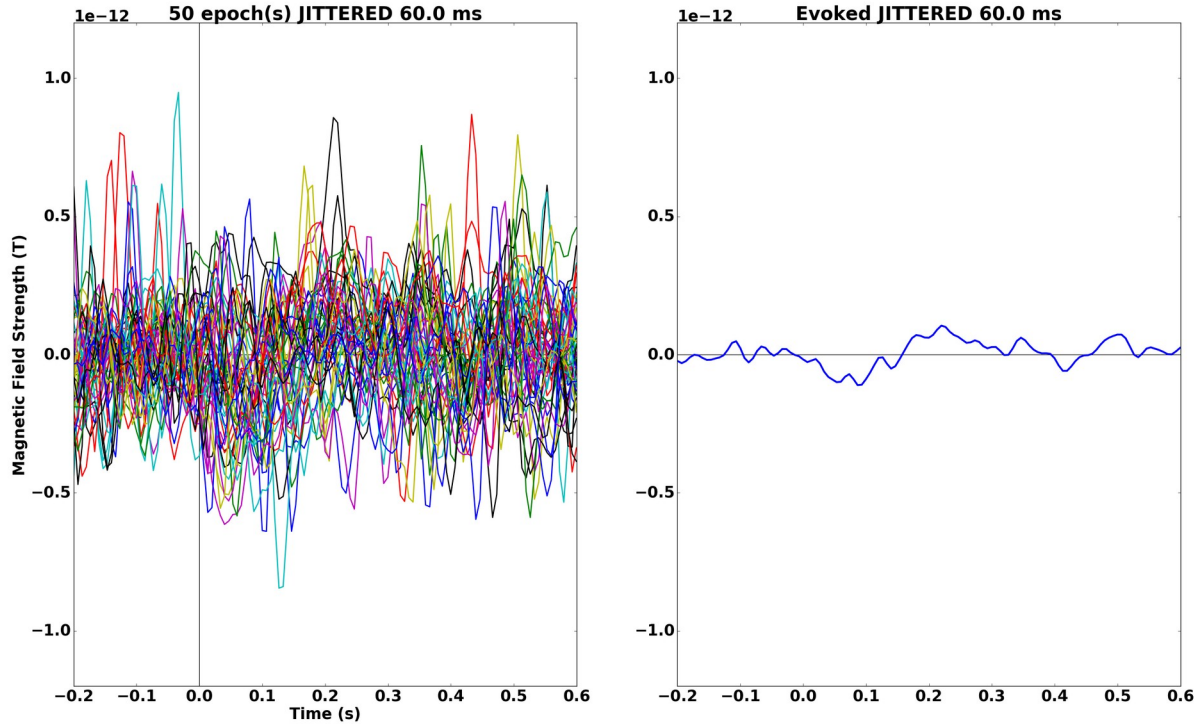
Epochs and Evoked JITTERED 60 ms



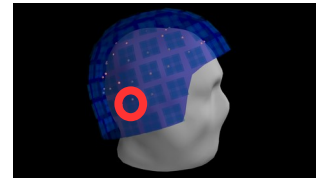
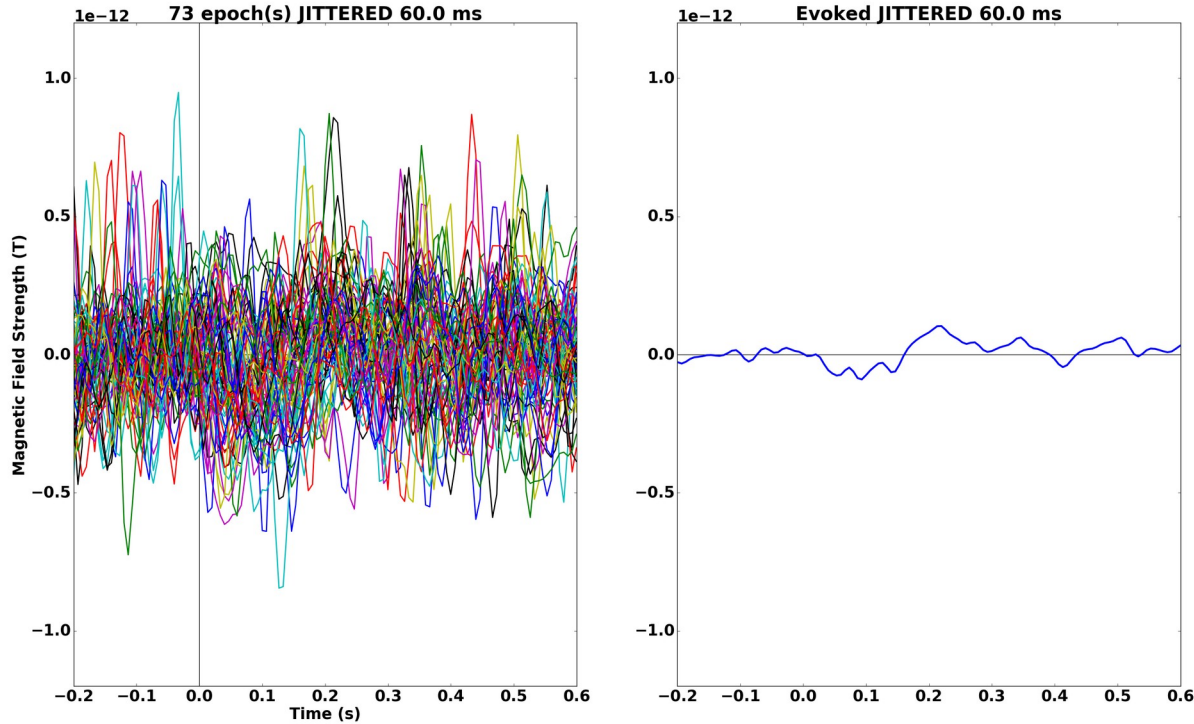
Epochs and Evoked JITTERED 60 ms



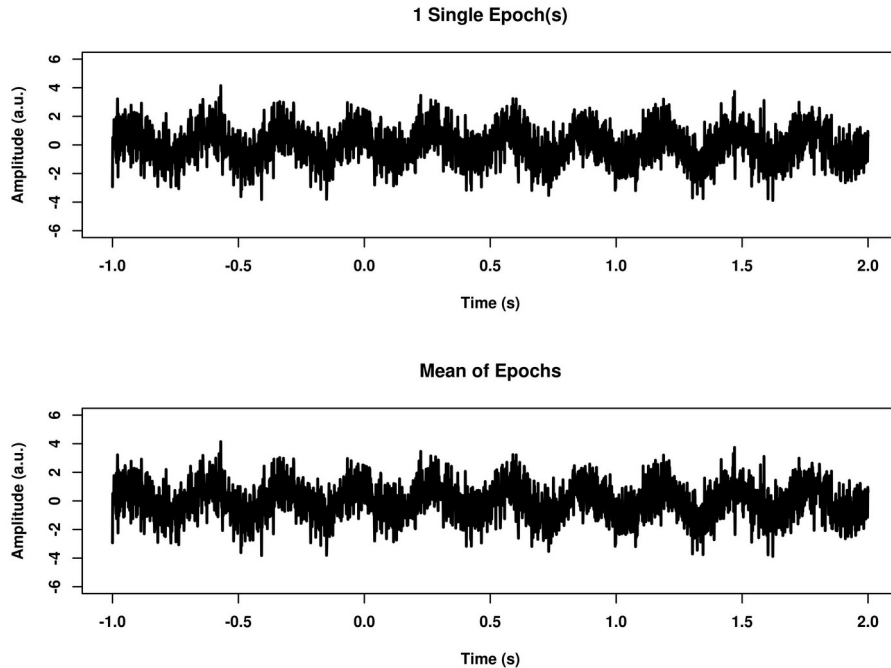
Epochs and Evoked JITTERED 60 ms



Epochs and Evoked JITTERED 60 ms

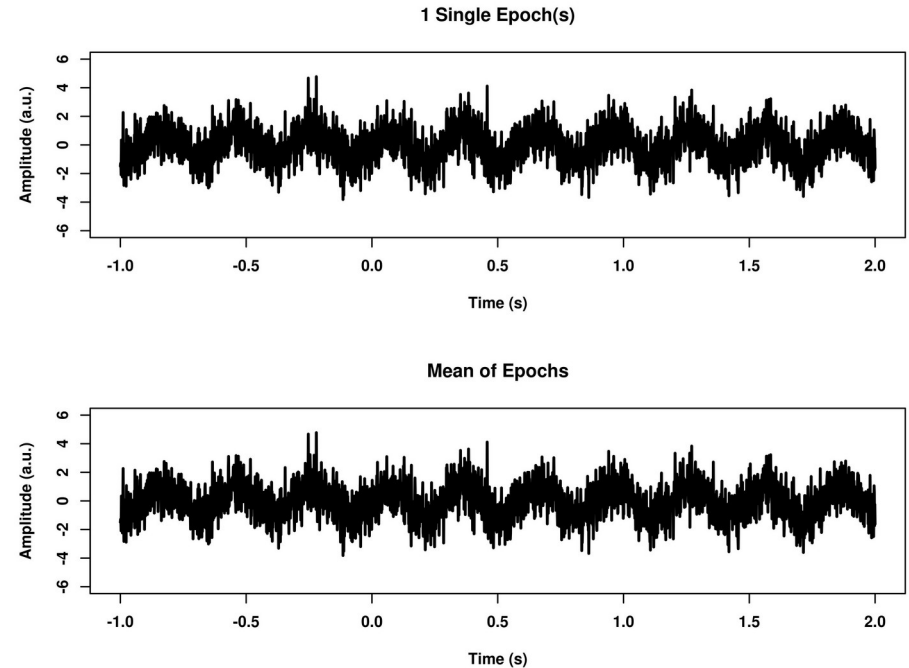


In-phase or out-of-phase with noise



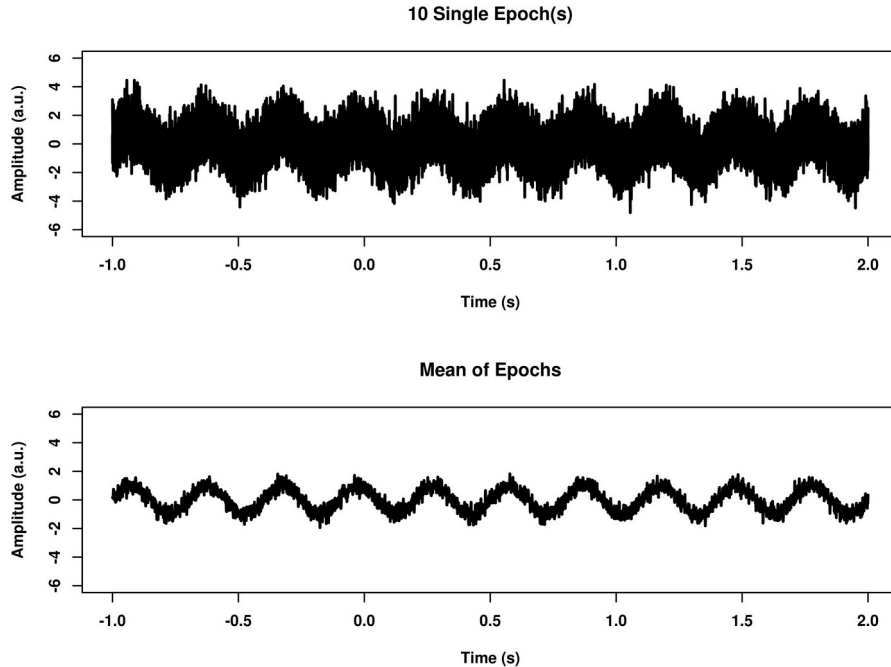
IN PHASE

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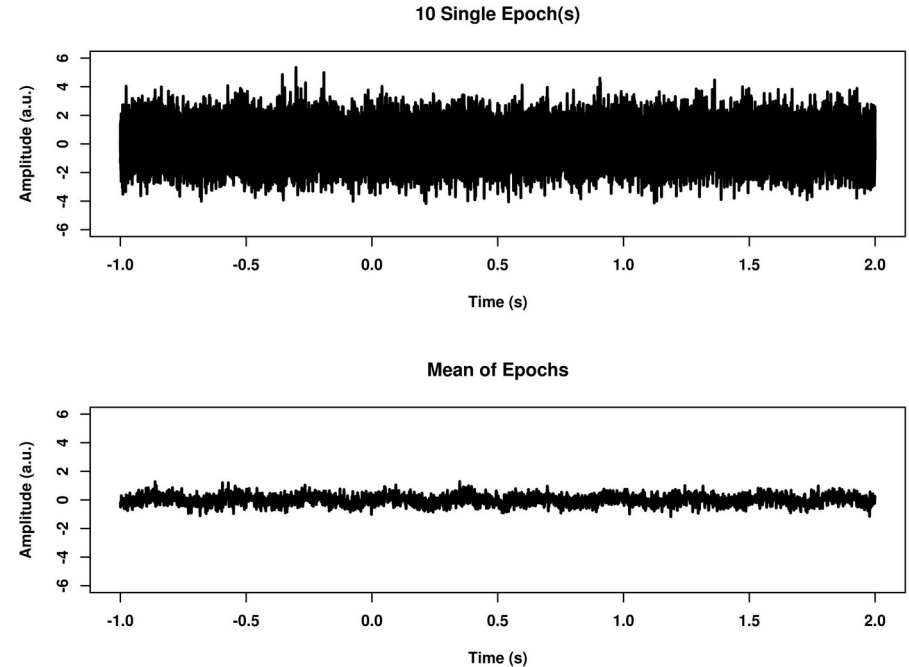


OUT OF PHASE

In-phase or out-of-phase with noise

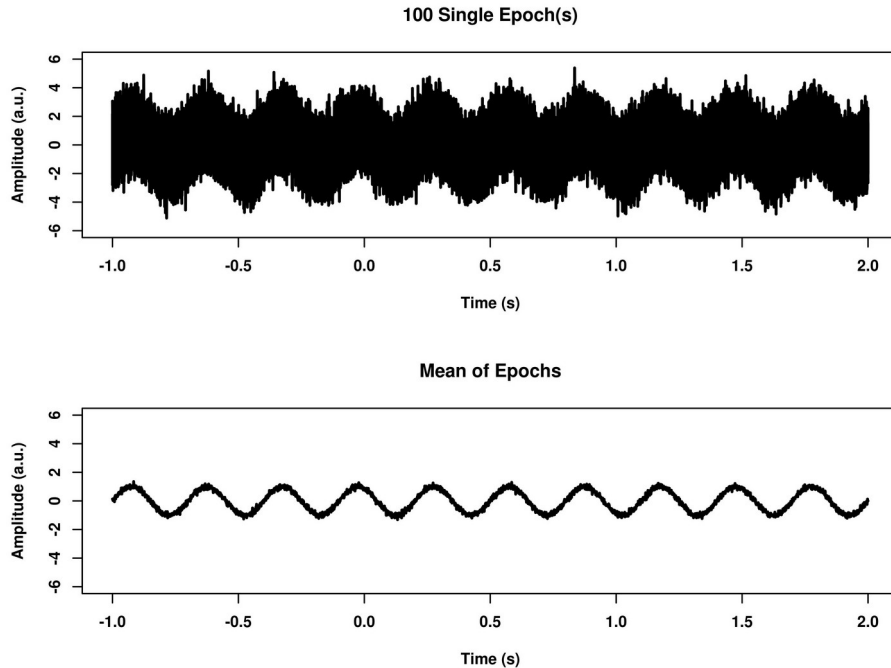


IN PHASE



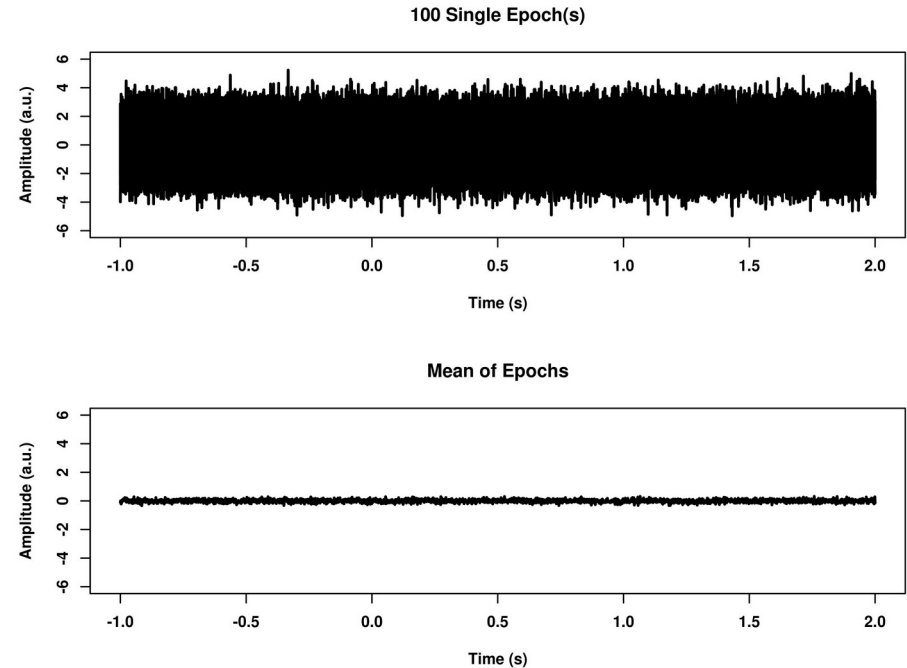
OUT OF PHASE

In-phase or out-of-phase with noise



IN PHASE

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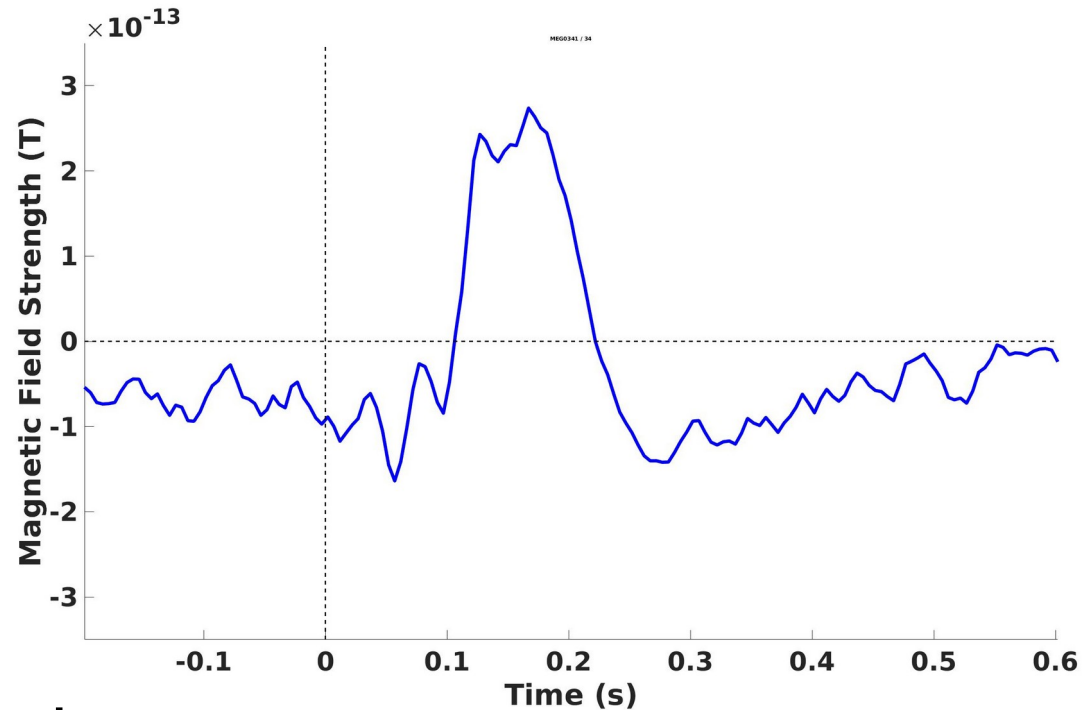


OUT OF PHASE

Demeaning (baselining)

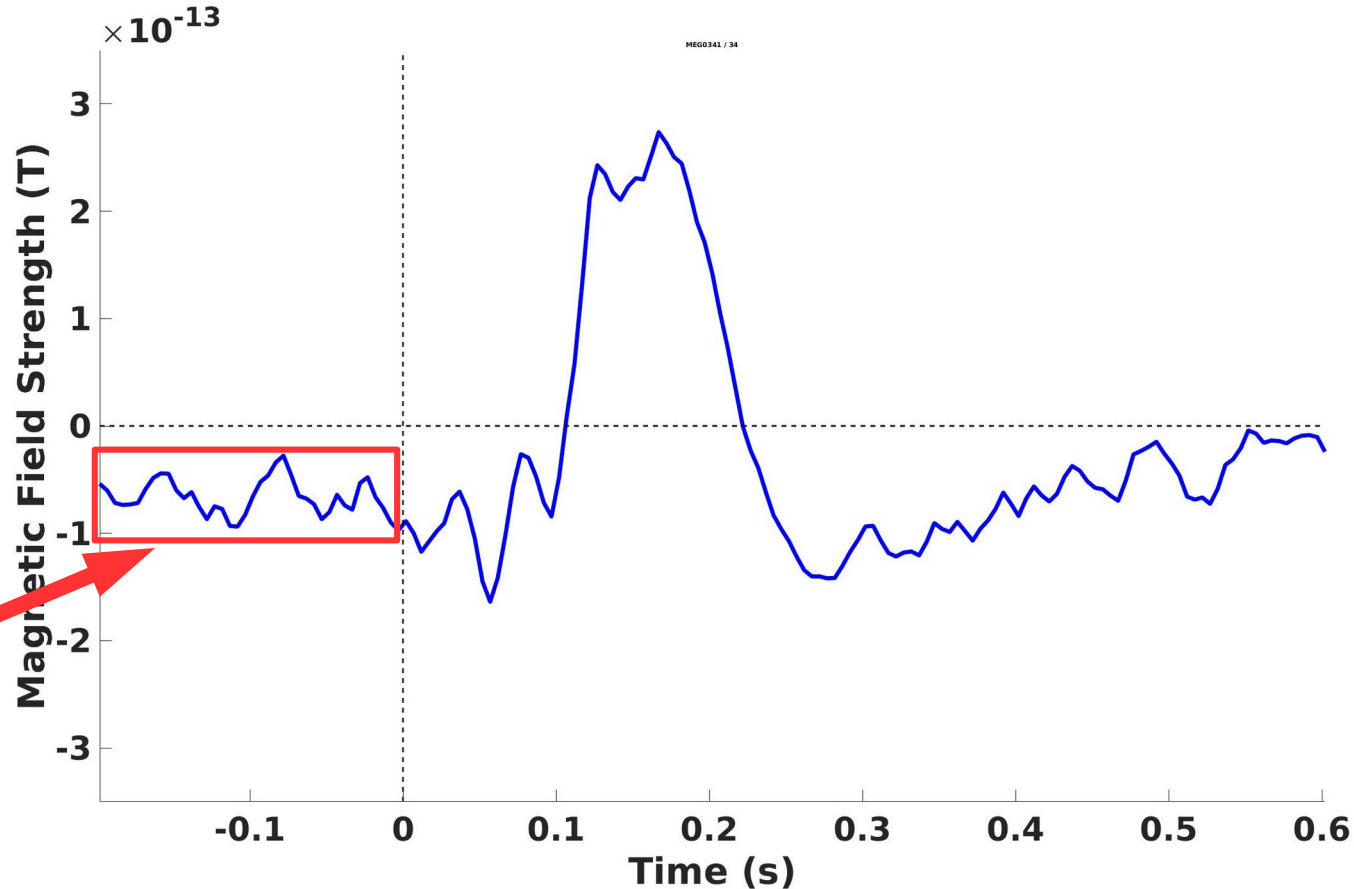
Note that all evoked responses
and epochs shown thus far
have been demeaned

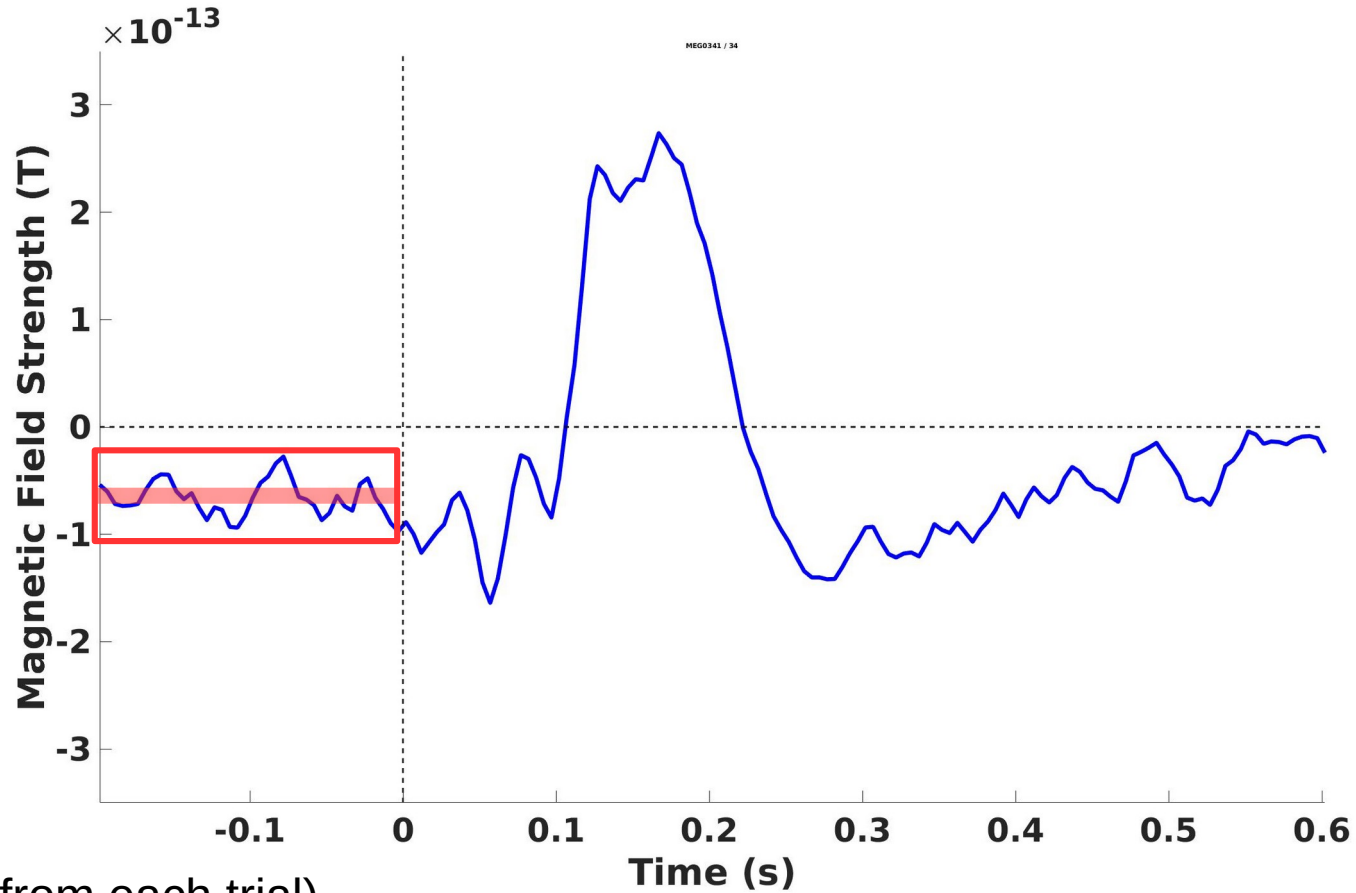
Demeaning (baselining)



Not demeaned yet –
how can we tell?

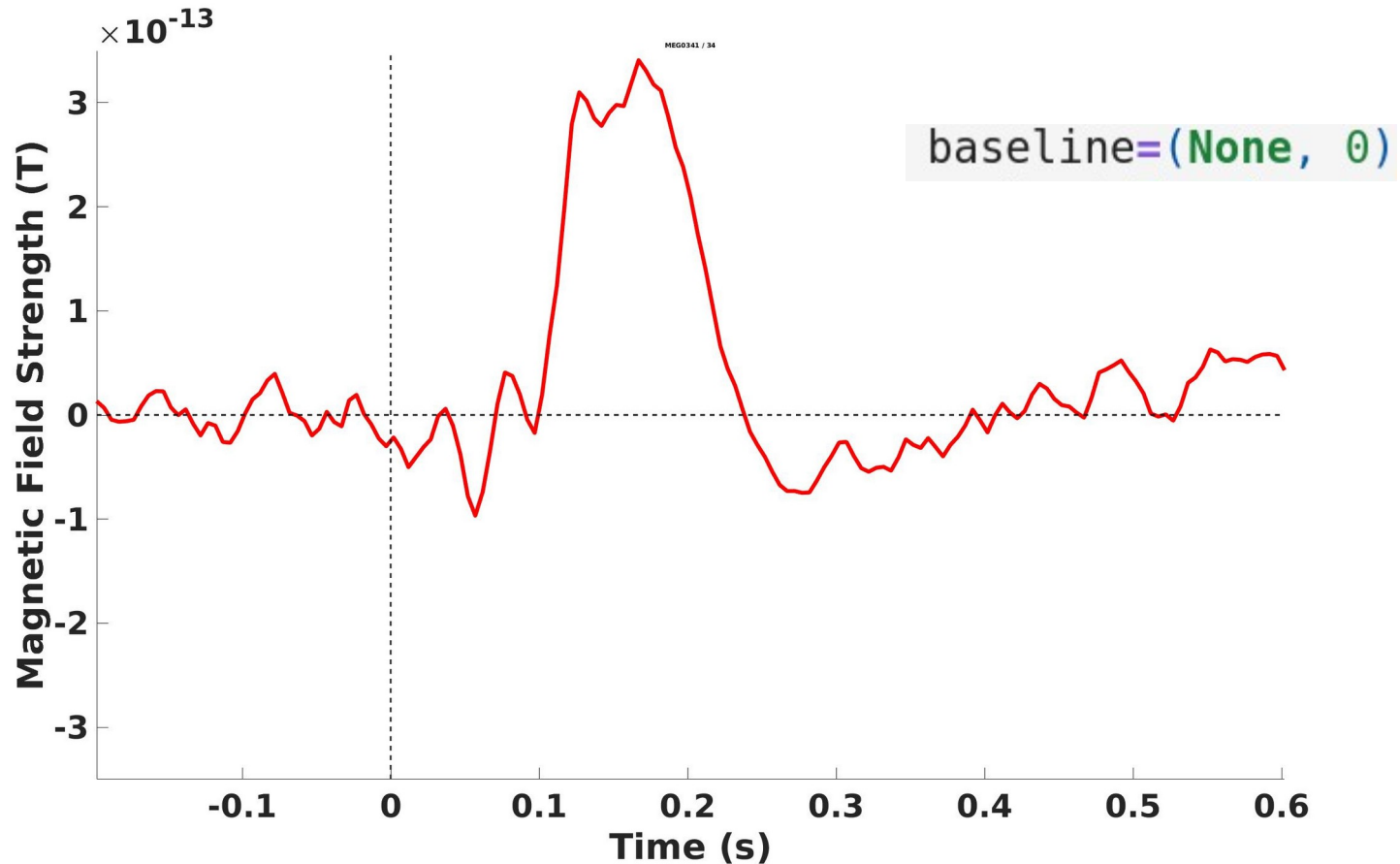
Demeaning (baselining)



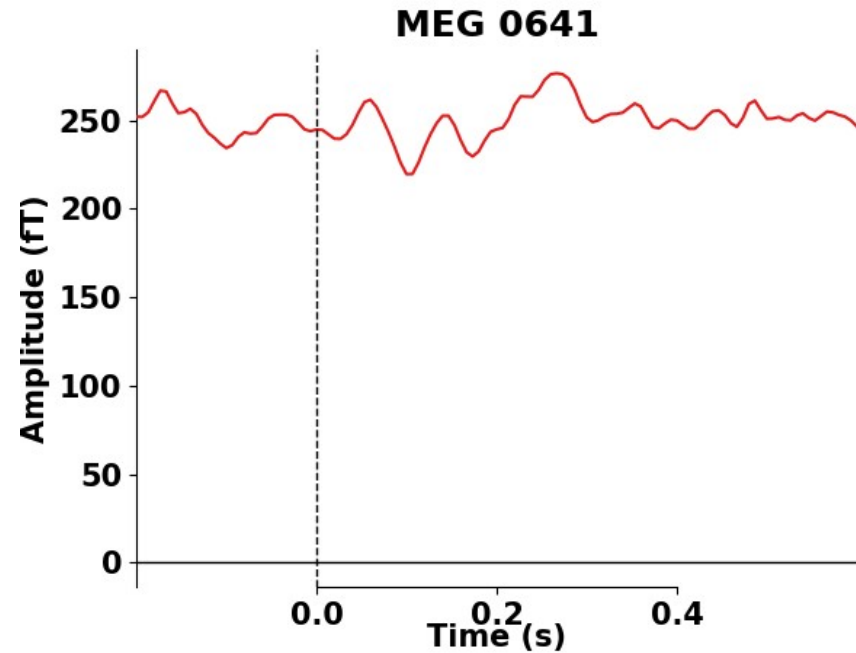
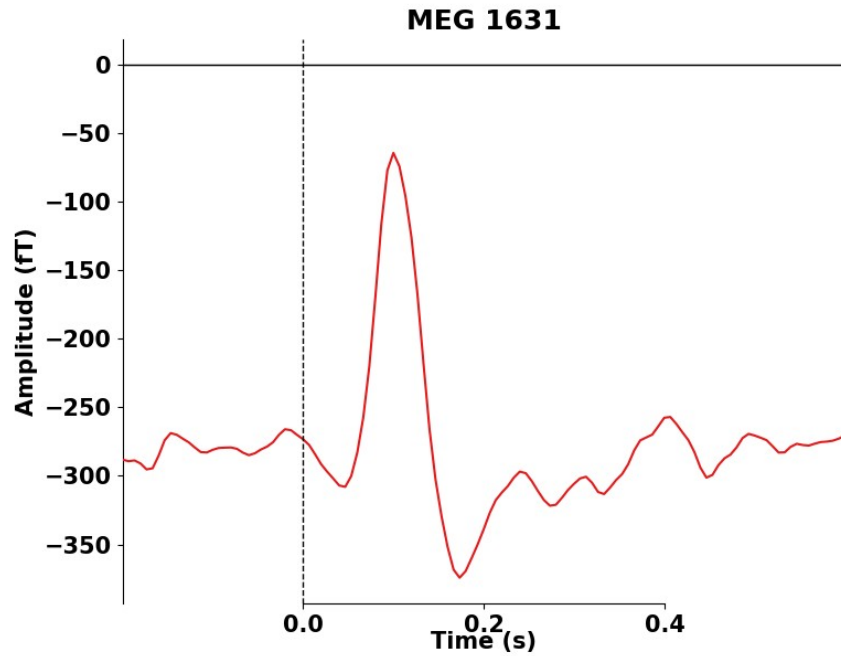


1. Find mean (from each trial)
2. Subtract the mean from each sample (time point)

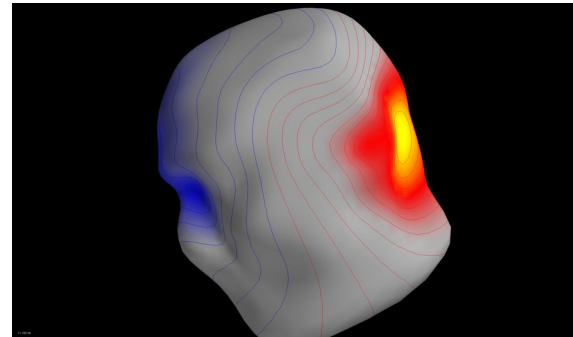
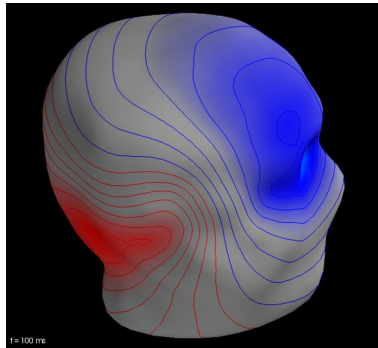
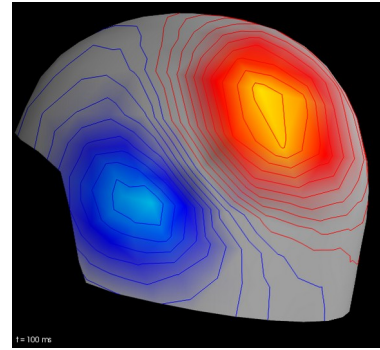
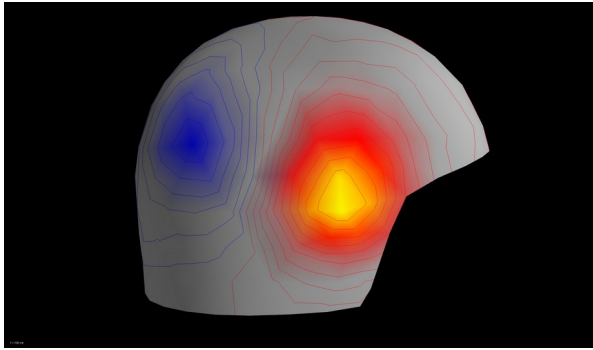
Demeaning (baselining)



Offsets may differ a lot

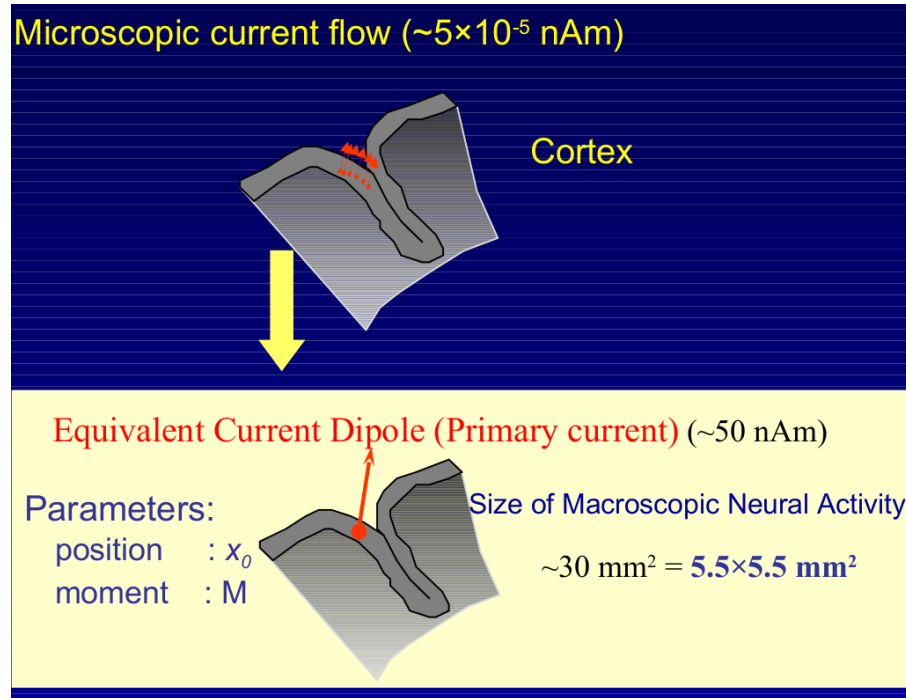


Field and potential plots; and localizing the source



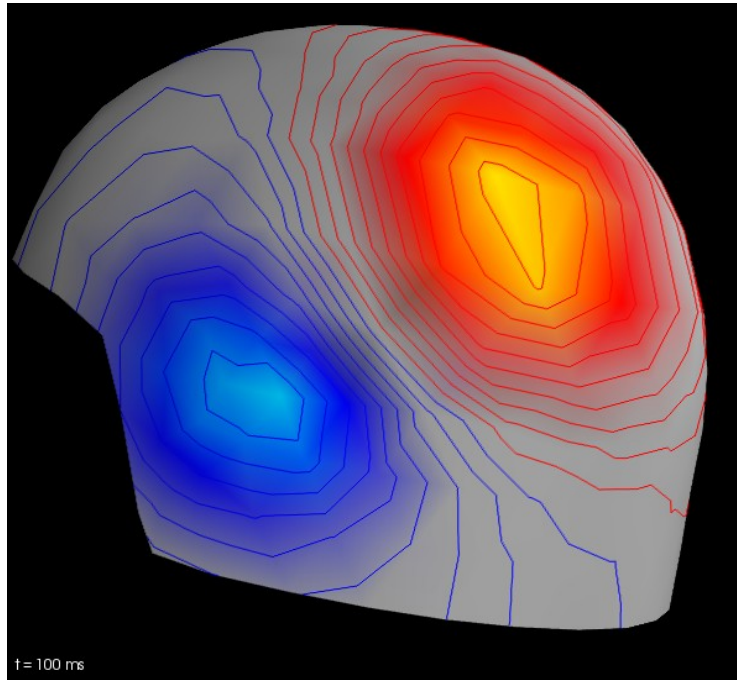
100 ms

Equivalent Current Dipole



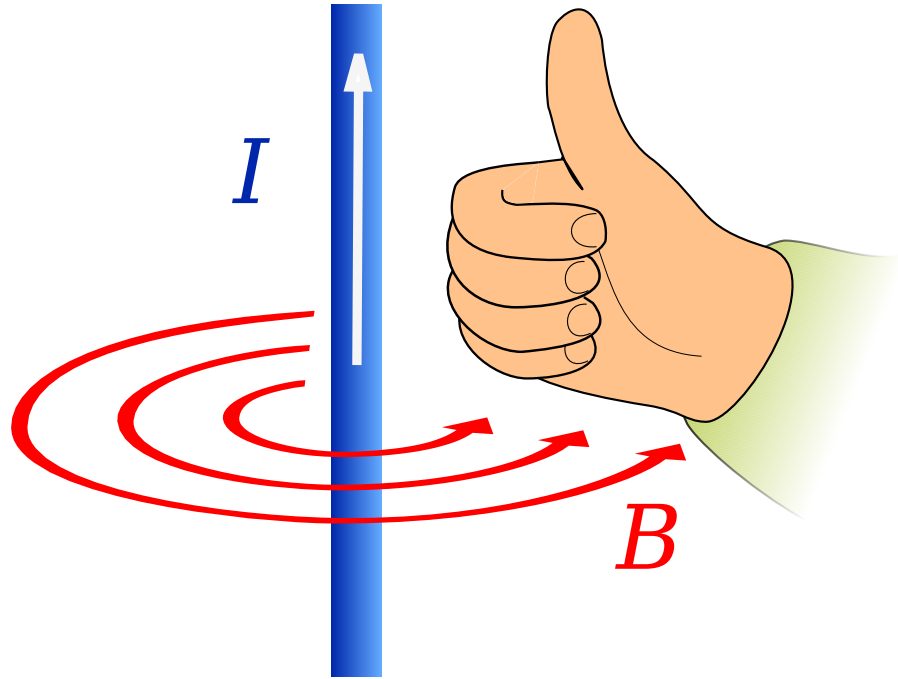
Stephanie Sillekens

Field plot

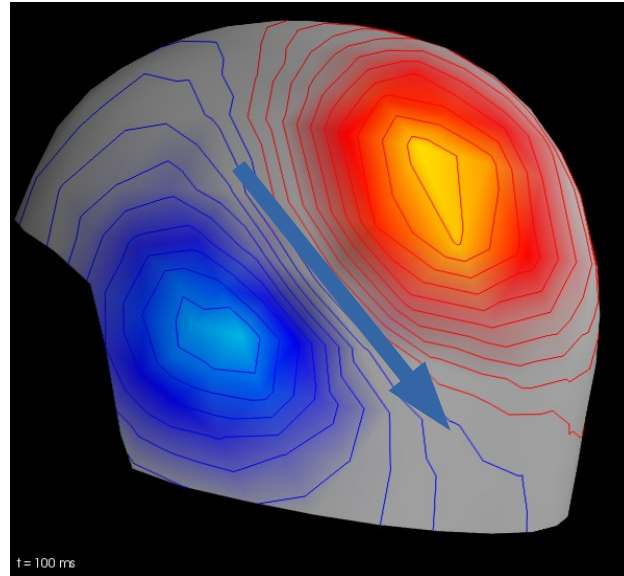


Which way does current flow in the underlying dipole?

Right-hand rule



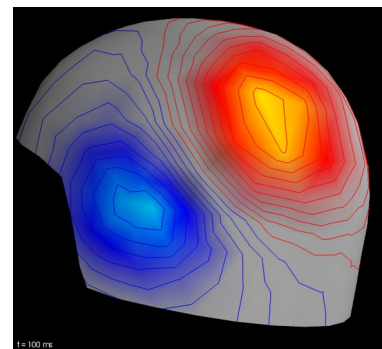
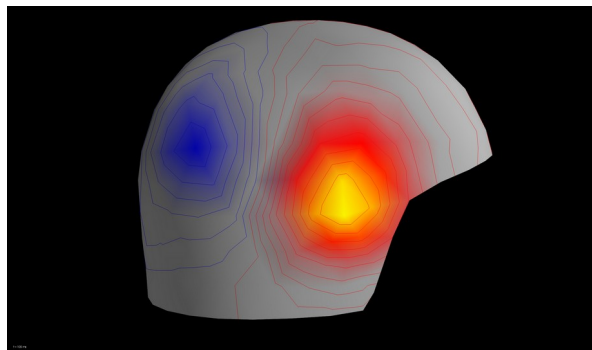
Field plot with very rough source localization



Evoked representations

Evoked representations 3d-topography

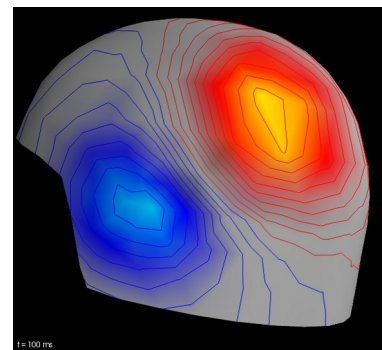
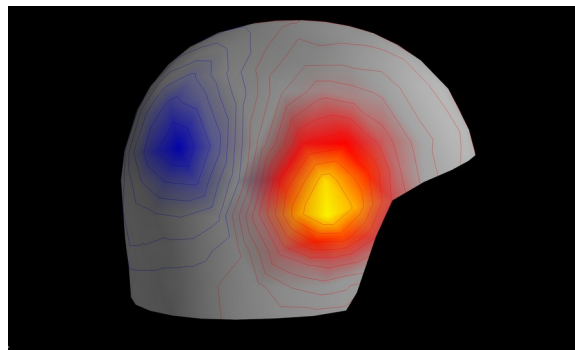
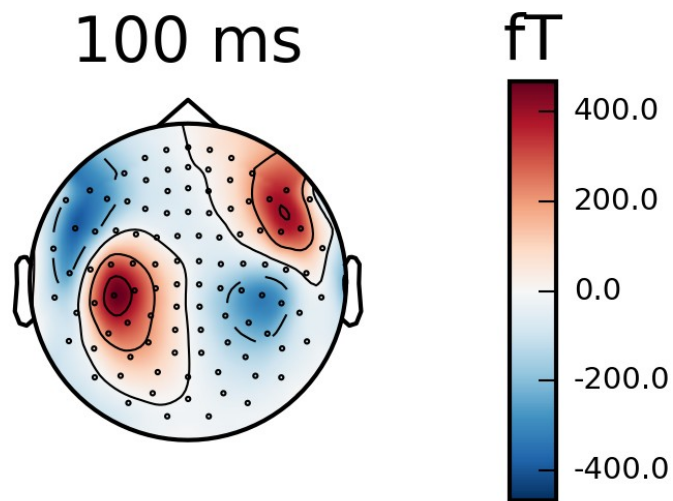
100 ms



Evoked representations

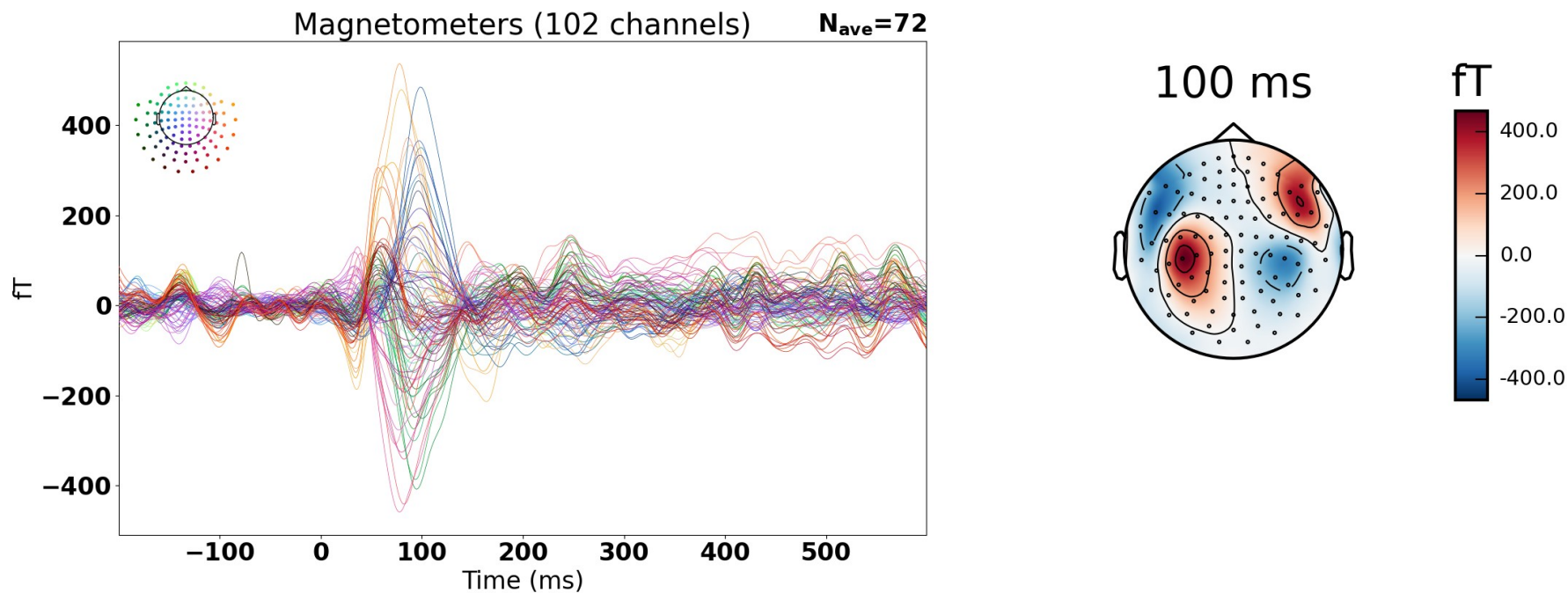
2d-topography

auditory_right mag



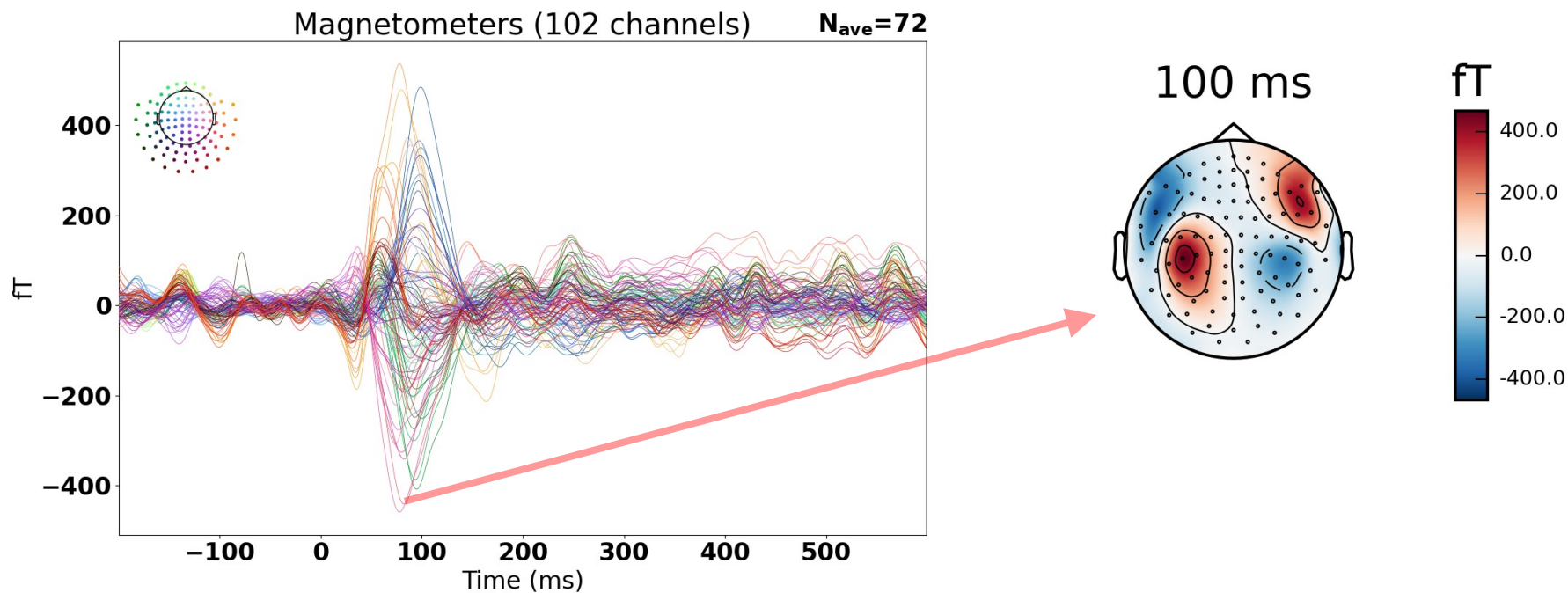
Evoked representations

Butterfly plot



Evoked representations

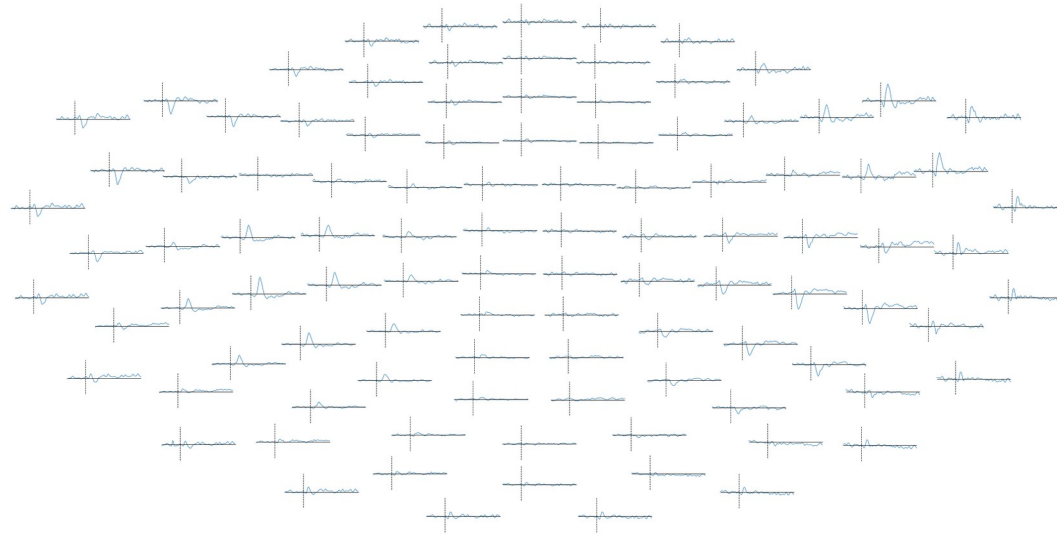
Butterfly plot



Evoked representations

Channel topography

LA



Learning goals

- Did you learn?
 - the identification of physiological artefacts
 - strategies for increasing the signal-to-noise ratio
 - how to use epochs to create an evoked response
 - the importance of epochs being in-phase for an averaging strategy to be feasible
 - to do rough source localization on dipolar activations

The course plan

Week 37:

Monday: Workshop paradigm: Measuring visual subjective experience
MR Recordings (Lau)

Week 38:

Monday: What is it all about? (Lau)
Workshop 1: MR-recordings (Laura & Lau)

Week 39:

Monday: Basic physiology and Evoked responses (Lau)
Wednesday: Preprocessing and evoked responses (Laura)

Week 40:

Monday: Forward modelling and inverse modelling (Lau)
Wednesday: Source reconstruction (Laura)

Week 41:

Monday: Multivariate statistics (Laura) (Lecture)
Wednesday: Multivariate statistics (Laura) (Coding)

Deadline for feedback: Video Explainer

Week 43:

Monday: Inverse modelling: beamforming (Lau) (Lecture)
Wednesday: Inverse modelling: beamforming (Laura) (Coding)

Week 44:

Monday From single subject analysis to group analysis (Laura)
Wednesday: Work on your reports (Laura)

Week 45:

Monday: What about that other cortex? - the cerebellar one (Lau)
Wednesday: Work on your reports, continued (Laura)

Week 46:

Monday: Optically pumped magnetoencephalography (Laura)
Wednesday: Respiration (Laura)

Deadline for feedback: Lab report

Week 47:

No teaching: prepare presentations!

Week 48:

Monday: Oral presentations (part 1) (Lau)
Wednesday: Oral presentations (part 2) (Lau)

Week 49:

Monday: Oral presentations (part 3) (Lau)
Wednesday: Oral presentations (part 4) (Lau)

Week 50:

Wednesday: What was it all about (Lau)